

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

Chap. 4 Open-Loop Discrete-Time Systems

Chap. 5 Closed-Loop Systems

Chap. 6 System Time-Response Characteristics

Chap. 7 Stability Analysis Techniques

Chap. 8 Digital Controller Design

6.1 INTRODUCTION

$$c(t) = \underline{c_{\text{forced}}(t)} + c_{\text{natural}}(t) \quad (6.1)$$

Steady-state resp.

- ❖ Stability Definitions (*For LTI Systems*)
 - Using the natural response (approaches zero)
 - Stable, unstable, and marginally stable
 - Using the total response (BIBO)
 - Stable and unstable (includes marginally stable)



- ❖ Goal: *Determine if a system is stable.*

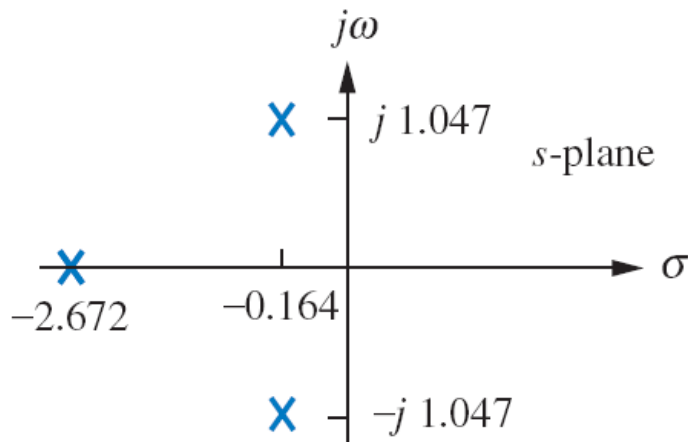
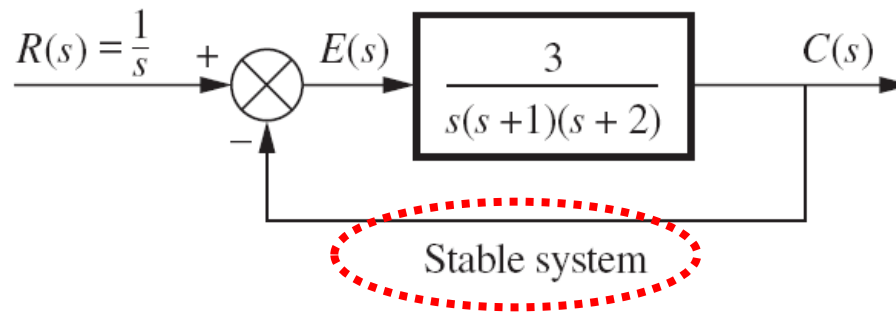
- ❖ Left/Right half-plane (lhp/rhp)

Unit multiplicity!

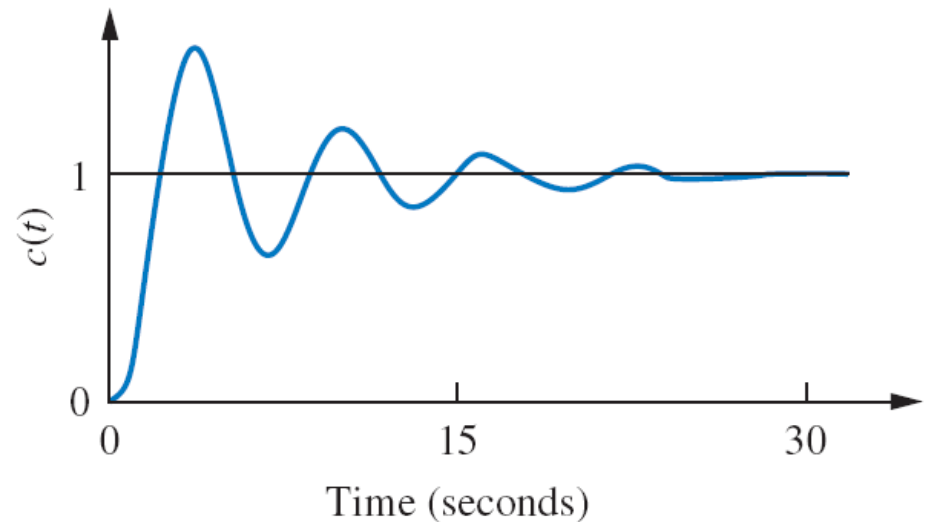
- ❖ Poles of multiplicity greater than 1 *on the imaginary axis* lead to $A t^n \cos(\omega t + \phi)$, where $n = 1, 2, \dots$, which also approaches infinity.

FIGURE 6.1 Closed-loop poles and response:

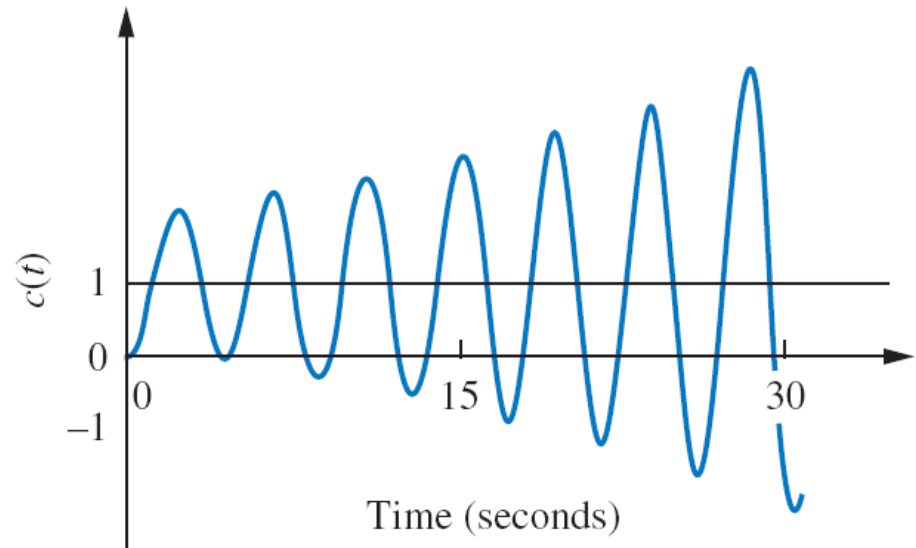
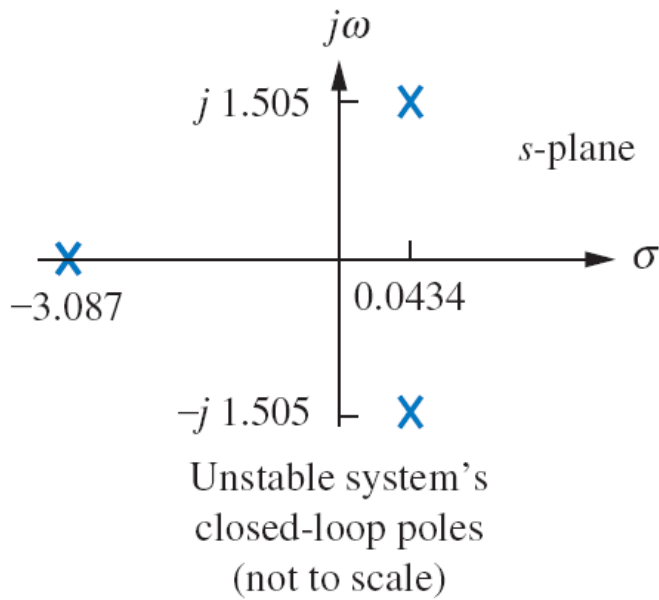
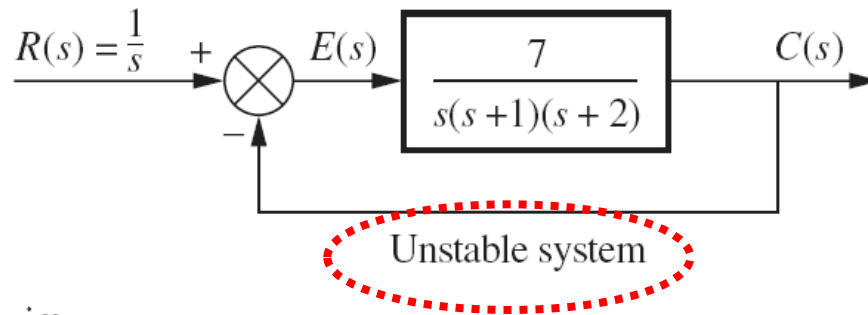
- a. stable system;
- b. unstable system



Stable system's
closed-loop poles
(not to scale)



(a)



(b)

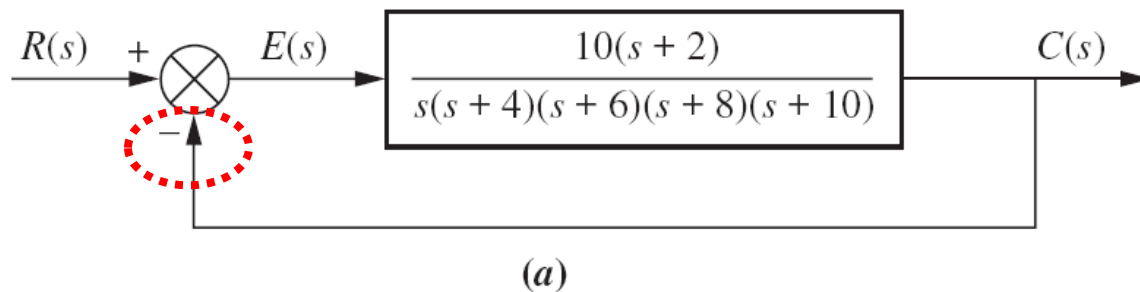
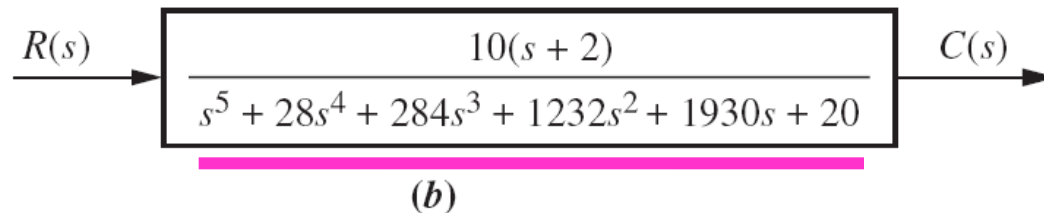


FIGURE 6.2 Common cause of problems in finding closed-loop poles:
a. original system;
b. equivalent system

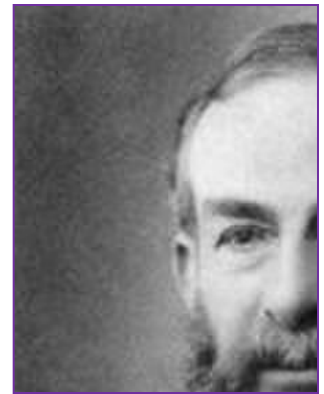


➡ All **positive** coefficients (the same signs)

➡ **Stable**

6.2 ROUTH-HURWITZ CRITERION

- R-H Criterion: Check stability without the need to solve for the closed-loop systems poles.
- Two Steps:
 - 1) Generate a *Routh Table*
 - 2) Interpret the Table
- The power of R-H criterion lies in design (range of K to yield stability) rather than analysis.



Prof. Routh
(1831-1907)

GENERATING A BASIC ROUTH TABLE

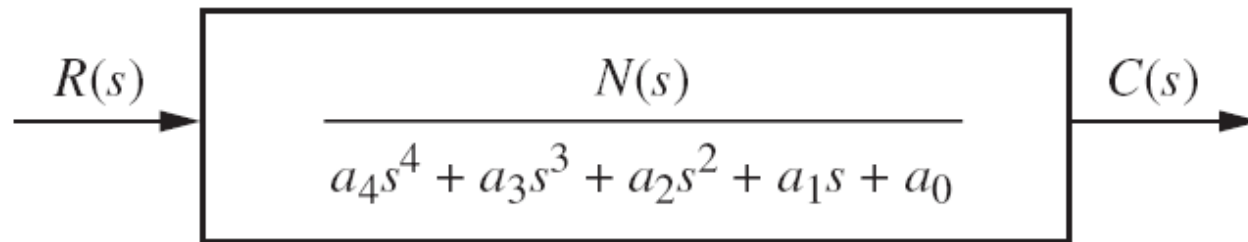


FIGURE 6.3 Equivalent closed-loop
transfer function

TABLE 6.1 Initial layout for Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2			
s^1			
s^0			

TABLE 6.2 Completed Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$-\frac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$	$-\frac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$	$-\frac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$
s^1	$-\frac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$	$-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$	$-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$
s^0	$-\frac{\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$	$-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$	$-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$

EXAMPLE 6.1

Creating a Routh table

Problem: Make the Routh table for the system shown in Figure 6.4(a).

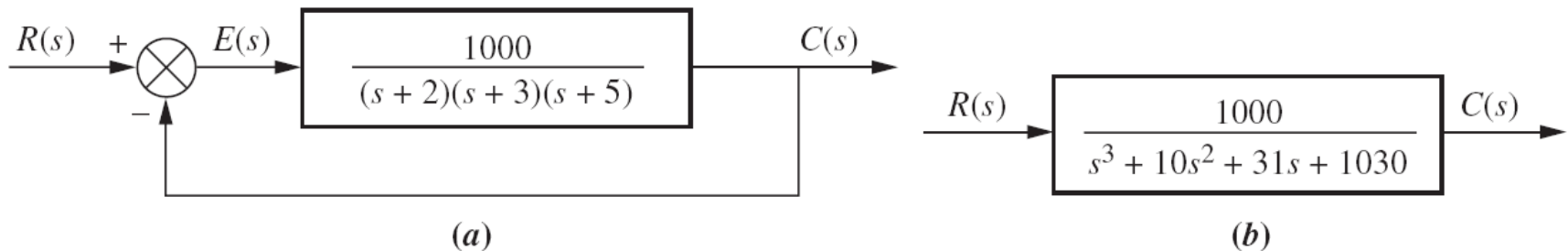


FIGURE 6.4 a. Feedback system for Example 6.1;
b. equivalent closed-loop system

EXAMPLE 6.1

TABLE 6.3 Completed Routh table for Example 6.1

s^3	1	31	0
s^2	10 1	1030 103	0
s^1	$\frac{-\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$	$\frac{-\begin{vmatrix} 1 & 0 \\ \cancel{0} 1 & 0 \end{vmatrix}}{1} = 0$	$\frac{-\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$
s^0	$\frac{-\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$	$\frac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$	$\frac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$

➡ *Unstable* (2 poles in the rhp).

INTERPRETING THE BASIC ROUTH TABLE

- The number of **roots in the rhp** is equal to the number of ***sign changes*** in the 1st column.
- ***Stable*** → no sign changes.
- Example 6.1: Unstable (2 poles in the rhp).



Prof. Hurwitz
(1859-1919)

6.3 ROUTH-HURWITZ CRITERION: SPECIAL CASES

- Two Special Cases
 - Zero Only In The First Column
 - Epsilon Method: ε approaches zero
 - Reverse Coefficients Method
 - Entire Row Is Zero
 - An even polynomial is a factor of the original polynomial
 - Form an auxiliary polynomial

7.2 Stability

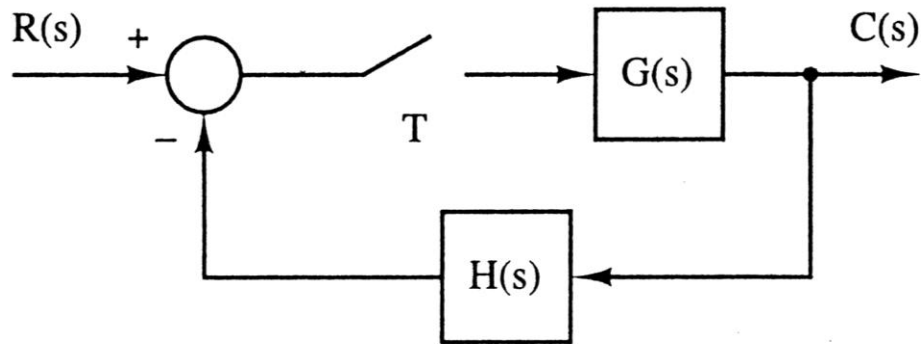


Figure 7-1 Sampled-data system.

$$C(z) = \frac{G(z)R(z)}{1 + \overline{GH}(z)} = \frac{K \prod_{i=1}^m (z - z_i)}{\prod_{i=1}^n (z - p_i)} R(z)$$

If distinct poles

$$C(z) = \frac{k_1 z}{z - p_1} + \dots + \frac{k_n z}{z - p_n} + C_R(z) \quad (7.1)$$

Natural/transient resp.

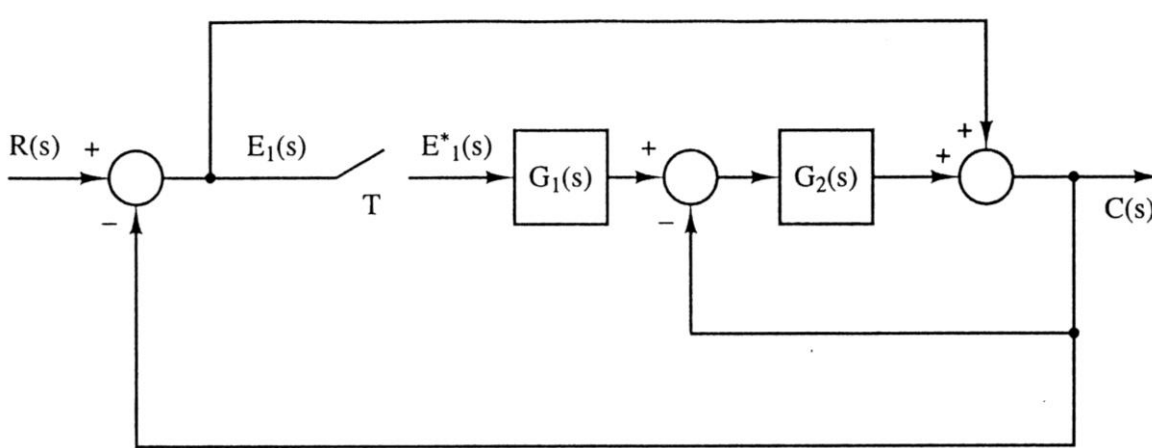
$$z^{-1} \left[\frac{k_i z}{z - p_i} \right] = k_i (p_i)^k \quad (7.2)$$

Thus, if the magnitude is less than 1, this term approach zero as k approaches infinity.

➡ Stable!

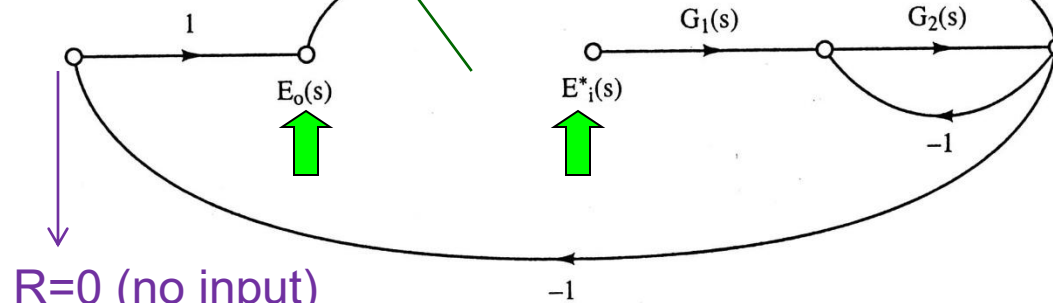
The characteristic equation is

$$1 + \overline{GH}(z) = 0 \quad (7.3)$$



(a)

Open the system
at the sampler



R=0 (no input)
Stability-independent

(b)

Figure 7-2 Discrete-time system.

方法1

From Ex 5.3 (P182)

$$C(z) = \left[\frac{R}{2 + G_2} \right](z) + \frac{\left[\frac{G_1 G_2}{2 + G_2} \right](z)}{1 + \left[\frac{G_1 G_2}{2 + G_2} \right](z)} \left[\frac{(1 + G_2)R}{2 + G_2} \right](z)$$

Char. Equations

方法2

$$E_0(s) = \frac{-G_1 G_2}{2 + G_2} E_i^*(s)$$

Z-transform

$$E_0(z) = - \left[\frac{G_1 G_2}{2 + G_2} \right](z) E_i(z)$$

Open-loop transfer function

$$G_{op}(z) = \frac{E_0(z)}{E_i(z)} = - \left[\frac{G_1 G_2}{2 + G_2} \right](z)$$

Close-loop system: $E_i(z) = E_0(z)$

$$[1 - G_{op}(z)] E_0(z) = 0$$

The characteristic equation

$$1 + \left[\frac{G_1 G_2}{2 + G_2} \right](z) = 0$$

Example 7.1

方法2

Consider the system of Example 5.2, which is repeated in Figure 7-3a. The flow graph of the system is opened at the first sampler as shown in Figure 7-3b. The effect of the second sampler is included, by denoting its input as $E_1(s)$ and its output as $E_1^*(s)$. From this flow graph we write

$$E_1 = G_1 E_i^* - G_2 H E_1^*$$

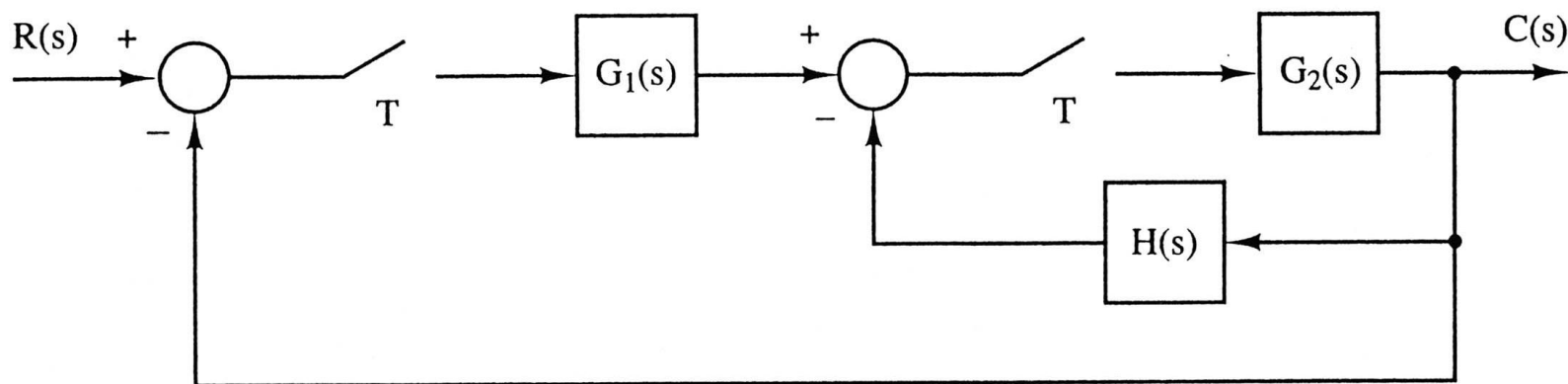
$$E_o = -G_2 E_1^*$$

Starring the first equation and solving for E_1^* , we obtain

$$E_1^* = \frac{G_1^* E_i^*}{1 + \overline{G_2 H^*}}$$

Starring the second equation and substituting in the value for E_1^* , we obtain

$$E_o^* = \frac{-G_1^* G_2^*}{1 + \overline{G_2 H^*}} E_i^*$$



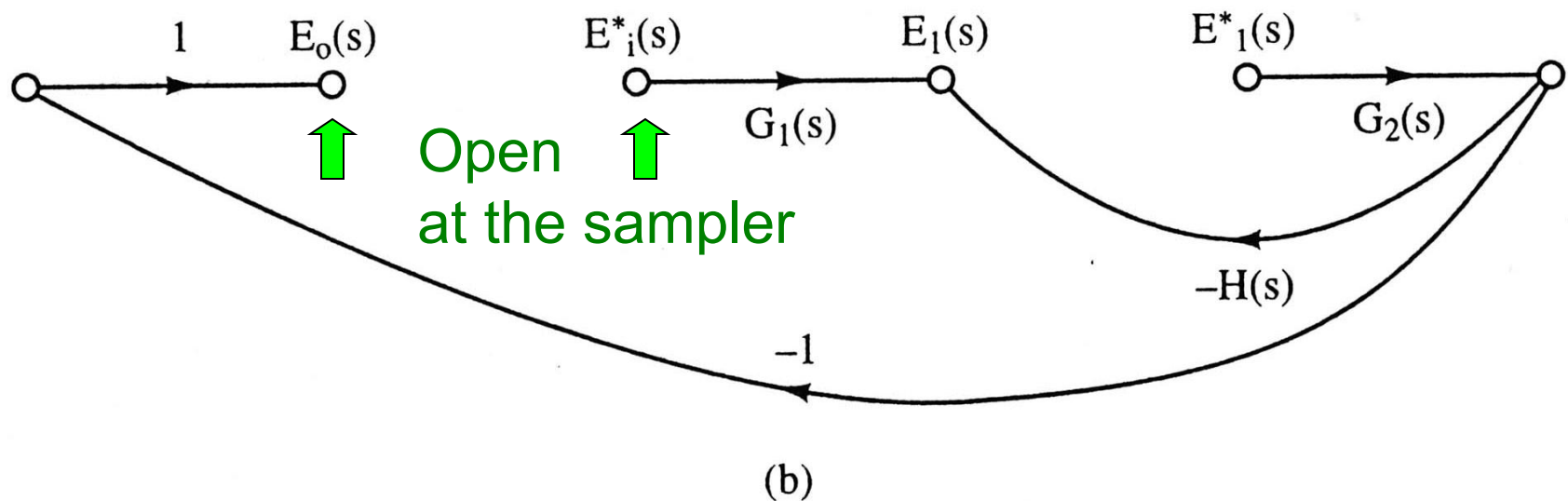


Figure 7-3 System for Example 7.1.

Since $E_i(z) = E_o(z)$ in the closed-loop system,

$$\left[1 + \frac{G_1(z)G_2(z)}{1 + \overline{G_2H}(z)} \right] E_o(z) = 0$$

Thus we can write the characteristic equation

(See P180) $\leftarrow 1 + G_1(z)G_2(z) + \overline{G_2H}(z) = 0$

This result is verified in Example 5.2. We leave the derivation of the characteristic equation by opening the system at the second sampler as an exercise for the reader (see Problem 7-2).

7.3 Bilinear Transformation

Use of the transform

$$z = \frac{1 + (T/2)w}{1 - (T/2)w} \quad (7.7)$$

or

$$w = \frac{2}{T} \frac{z - 1}{z + 1} \quad (7.8)$$

$j\omega$ vs. $j\omega_w$

On the unit circle in the z -plane

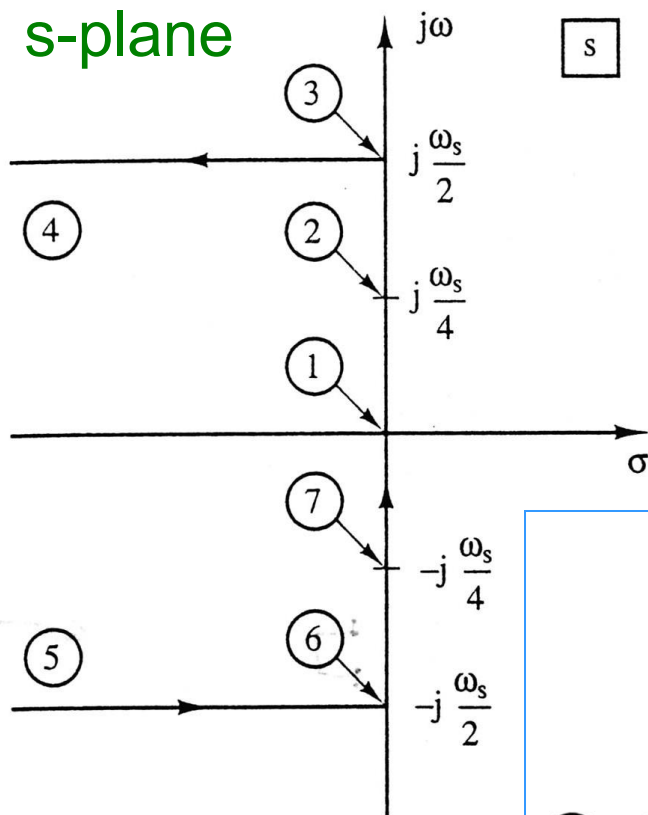
$$z = e^{j\omega T}$$

$$\begin{aligned} w &= \left. \frac{2}{T} \frac{z - 1}{z + 1} \right|_{z=e^{j\omega T}} = \frac{2}{T} \frac{e^{j\omega T} - 1}{e^{j\omega T} + 1} = \frac{2}{T} \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{e^{j\omega T/2} + e^{-j\omega T/2}} = j \frac{2}{T} \tan \frac{\omega T}{2} \\ &= j\omega_w \end{aligned} \quad (7.9)$$

Let $j\omega_w$ be the imaginary part of w . We will refer to ω_w as the w -plane frequency. Then can be expressed as

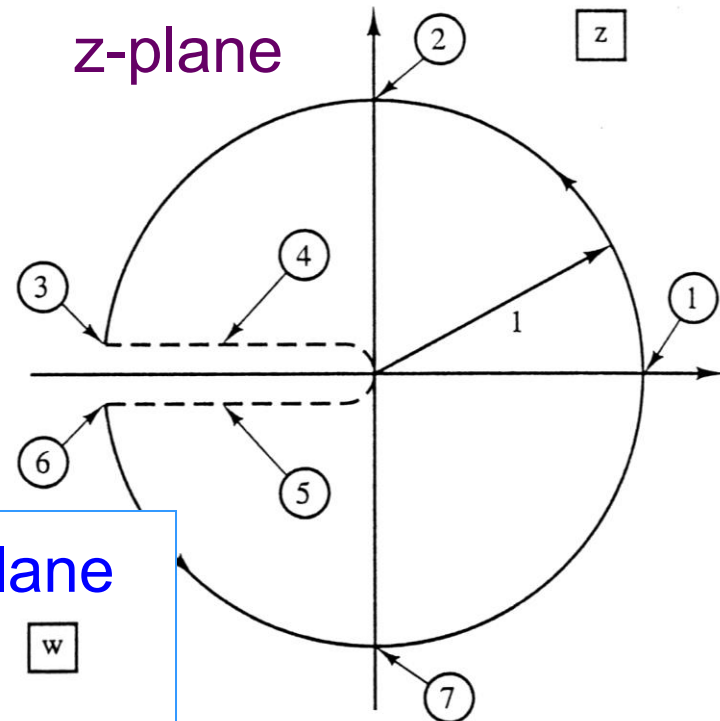
$$\omega_w = \frac{2}{T} \tan \frac{\omega T}{2} \quad (7.10)$$

s-plane



$$Z = e^{sT}$$

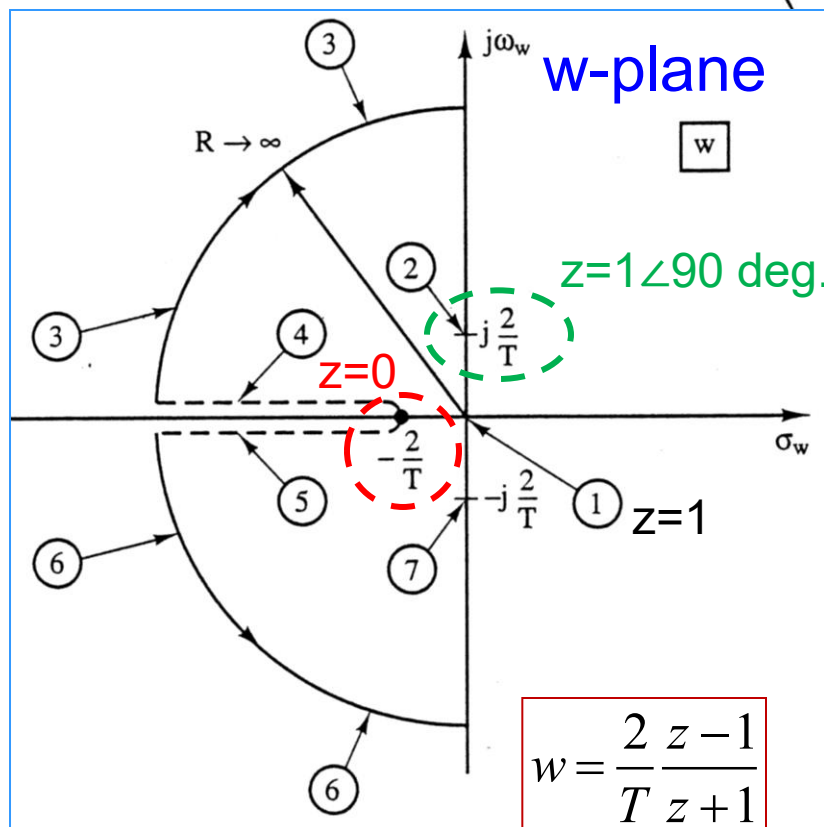
z-plane



$j\omega$ vs. $j\omega_w$

$$\omega_w = \frac{2}{T} \tan \frac{\omega T}{2}$$

w-plane



$$z = \frac{1 + (T/2)w}{1 - (T/2)w}$$

$$w = \frac{2}{T} \frac{z-1}{z+1}$$

Figure 7-4 Mapping from s-plane to z-plane to w-plane.

$j\omega$ vs. $j\omega_w$

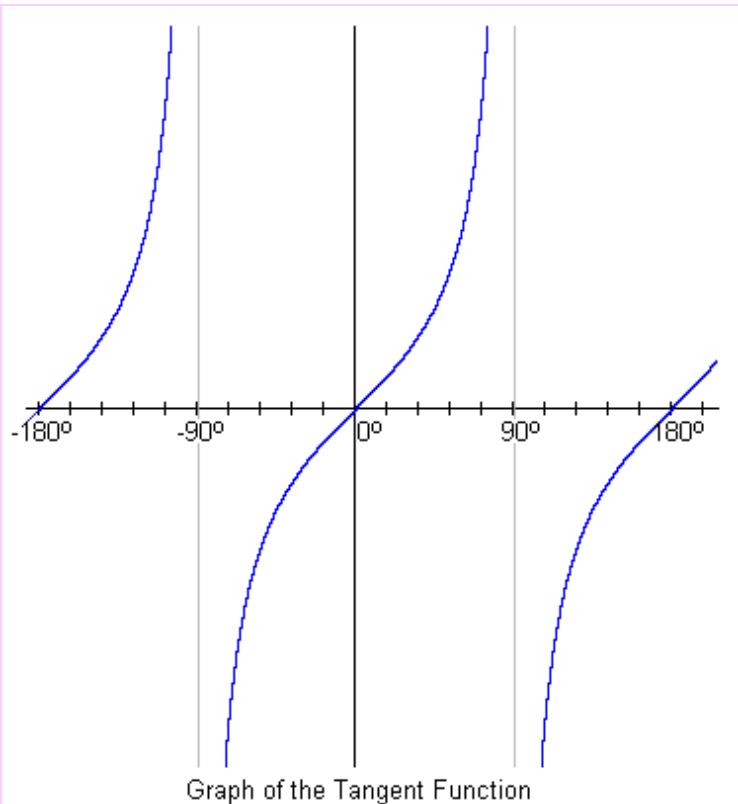
Tangent Fcn.

For small value of ωT ,

$$\Rightarrow \omega_w = \frac{2}{T} \tan \frac{\omega T}{2} \doteq \omega \quad (7.11)$$

$$\frac{\omega T}{2} \leq \frac{\pi}{10} \Rightarrow \omega \leq \frac{2\pi}{10T} = \frac{\omega_s}{10} \quad (7.12)$$

- The error in this approximation is less than 4 percent.
- As $\omega = \omega_s/10$, the ZOH's phase lag = 18 deg. → **affect stability.**



7.4 The Routh-Hurwitz Criterion

TABLE 7-1 BASIC PROCEDURE FOR APPLYING THE ROUTH-HURWITZ CRITERION

1. Given a characteristic equation of the form

$$F(w) = b_n w^n + b_{n-1} w^{n-1} + \cdots + b_1 w + b_0 = 0$$

form the Routh array as

$$\begin{array}{c|cccc} w^n & b_n & b_{n-2} & b_{n-4} & \cdots \\ w^{n-1} & b_{n-1} & b_{n-3} & b_{n-5} & \cdots \\ w^{n-2} & c_1 & c_2 & c_3 & \cdots \\ \vdots & d_1 & d_2 & d_3 & \cdots \\ w^1 & j_1 & & & \\ w^0 & k_1 & & & \end{array}$$

2. Only the first two rows of the array are obtained from the characteristic equation. The remaining rows are calculated as follows.

$$c_1 = \frac{b_{n-1} b_{n-2} - b_n b_{n-3}}{b_{n-1}}$$

$$d_1 = \frac{c_1 b_{n-3} - b_{n-1} c_2}{c_1}$$

$$c_2 = \frac{b_{n-1} b_{n-4} - b_n b_{n-5}}{b_{n-1}}$$

$$d_2 = \frac{c_1 b_{n-5} - b_{n-1} c_3}{c_1}$$

$$c_3 = \frac{b_{n-1} b_{n-6} - b_n b_{n-7}}{b_{n-1}}$$

$\vdots$

Why sign changes?

3. Once the array has been formed, the Routh-Hurwitz criterion states that the number of roots of the characteristic equation with positive real parts is equal to the number of sign changes of the coefficients in the first column of the array.
4. Suppose that the w^{i-1} th row contains only zeros, and that the w^i th row directly above it has the coefficients $\alpha_1, \alpha_2, \dots$. The auxiliary equation is then

$$\alpha_1 w^i + \alpha_2 w^{i-2} + \alpha_3 w^{i-4} + \cdots = 0 \rightarrow \text{Even polynomials}$$

This equation is a factor of the characteristic equation.

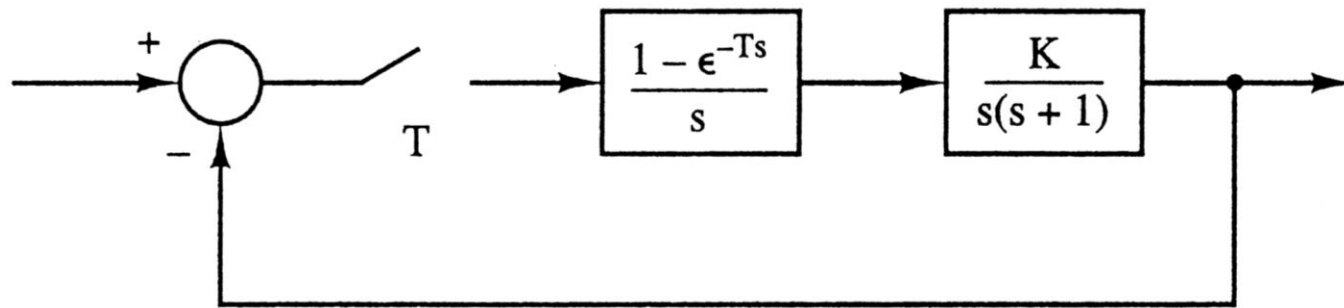


Figure 7-5 System for Examples 7.2 and 7.3.

Example 7.2

Consider the system shown in Figure 7-5, with $T = 0.1$ s. The open-loop function is

$$G(s) = \frac{1 - e^{-Ts}}{s} \left[\frac{1}{s(s+1)} \right]$$

Hence, from the z-transform tables we obtain

$$\begin{aligned} G(z) &= \frac{z-1}{z} \left[\frac{(\epsilon^{-T} + T - 1)z^2 + (1 - \epsilon^{-T} - T\epsilon^{-T})z}{(z-1)^2(z - \epsilon^{-T})} \right] \\ &= \frac{0.00484z + 0.00468}{(z-1)(z-0.905)} \end{aligned}$$

Then $G(w)$ is given by

$$\underline{G(w) = G(z)|_{z = [1 + (T/2)w]/[1 - (T/2)w]} = G(z)|_{z = (1 + 0.05w)/(1 - 0.05w)}}$$

or

$$G(w) = \frac{-0.00016w^2 - 0.1872w + 3.81}{3.81w^2 + 3.80w}$$

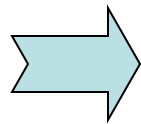
$$1 + \overline{GH}(z) = 0 \quad (7.3)$$

Then the characteristic equation is given by

$$1 + KG(w) = (3.81 - 0.00016K)w^2 + (3.80 - 0.1872K)w + 3.81 K = 0$$

The Routh array derived from this equation is

w^2	$3.81 - 0.00016K$	$3.81K$	$\Rightarrow K < 23,813$
w^1	$3.80 - 0.1872K$		$\Rightarrow K < 20.3$
w^0	$3.81K$		$\Rightarrow K > 0$



$$0 < K < 20.3$$

Stable!

Example 7.3

Consider again the system of Example 6.4 (shown again in Figure 7-5 with $T = 1$ s) with a gain factor K added to the plant. The characteristic equation is given by

$$\begin{aligned} 1 + KG(w) &= 1 + KG(z) \Big|_{z = (1 + 0.5w)/(1 - 0.5w)} \\ &= 1 + \frac{K \left[0.368 \left[\frac{1 + 0.5w}{1 - 0.5w} \right] + 0.264 \right]}{\left[\frac{1 + 0.5w}{1 - 0.5w} \right]^2 - 1.368 \left[\frac{1 + 0.5w}{1 - 0.5w} \right] + 0.368} \end{aligned}$$

or

$$\begin{aligned} 1 + KG(w) &= 1 + \frac{-0.0381K(w - 2)(w + 12.14)}{w(w + 0.924)} \\ &= \frac{(1 - 0.0381K)w^2 + (0.924 - 0.386K)w + 0.924K}{w(w + 0.924)} \end{aligned}$$

Thus the characteristic equation may be expressed as

$$(1 - 0.0381K)w^2 + (0.924 - 0.386K)w + 0.924K = 0$$

The Routh array is then

$$\begin{array}{l|ll} w^2 & 1 - 0.0381K & 0.924K \\ w^1 & 0.924 - 0.386K & \\ w^0 & 0.924K & \end{array} \Rightarrow \begin{array}{l} K < 26.2 \\ K < 2.39 \\ K > 0 \end{array}$$

Hence the system is stable for $0 < K < 2.39$.

*Degradation of stability as $T \uparrow$
is due to the delay (phase lag).*

7.5 Jury's Stability Test

Jury's test will now be presented. Let the characteristic equation of a discrete-time system be expressed as

$$\underline{Q(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0,} \quad a_n > 0 \quad (7-13)$$

Then form the array as shown in Table 7-2. Note that the elements of each of the

TABLE 7-2 ARRAY FOR JURY'S STABILITY TEST

z^0	z^1	z^2	$\dots$	z^{n-k}	$\dots$	z^{n-1}	z^n
a_0	a_1	a_2	$\dots$	a_{n-k}	$\dots$	a_{n-1}	a_n
a_n	a_{n-1}	a_{n-2}	$\dots$	a_k	$\dots$	a_1	a_0
b_0	b_1	b_2	$\dots$	b_{n-k}	$\dots$	b_{n-1}	
b_{n-1}	b_{n-2}	b_{n-3}	$\dots$	b_{k-1}	$\dots$	b_0	
c_0	c_1	c_2	$\dots$	c_{n-k}	$\dots$		
c_{n-2}	c_{n-3}	c_{n-4}	$\dots$	c_{k-2}	$\dots$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$			
l_0	l_1	l_2	l_3				
l_3	l_2	l_1	l_0				
m_0	m_1	m_2					

in reverse order

$$b_k = \begin{vmatrix} a_0 & a_{n-k} \\ a_n & a_k \end{vmatrix}, \quad c_k = \begin{vmatrix} b_0 & b_{n-1-k} \\ b_{n-1} & b_k \end{vmatrix}$$

$$d_k = \begin{vmatrix} c_0 & c_{n-2-k} \\ c_{n-2} & c_k \end{vmatrix} \dots$$
(7-14)

The necessary and sufficient conditions for the polynomial $Q(z)$ to have no roots outside or on the unit circle, with $a_n > 0$, are as follows:

$$\left. \begin{aligned} Q(1) &> 0 \\ (-1)^n Q(-1) &> 0 \\ |a_0| &< a_n \end{aligned} \right\} \begin{array}{l} \text{2nd-order systems.} \\ \text{w/o calculations.} \end{array}$$

$$\begin{aligned} |b_0| &> |b_{n-1}| \\ |c_0| &> |c_{n-2}| \\ |d_0| &> |d_{n-3}| \\ &\vdots \\ |m_0| &> |m_2| \end{aligned}$$
(7-15)

Note that for a second-order system, the array contains only one row. For each additional order, two additional rows are added to the array. Note also that for an n th-order system, there are a total of $n + 1$ constraints.

2nd-order → 3 constraints

Jury's test may be applied in the following manner :

1. Check the three conditions $Q(1) > 0, (-1)^n Q(-1) > 0$ and $|a_0| < a_n$, **which requires no calculations**. Stop if any of these conditions are not satisfied.
2. Construct the array, checking the conditions of (7-15) as each row is calculated. Stop if any condition is not satisfied.

2nd-order systems

Example 7.4

Consider again the system of Example 6.4 (and Example 7.3). Suppose that a gain factor K is added to the plant, and it is desired to determine the range of K for which the system is stable. Now, from Example 6.4, the system characteristic equation is

$$1 + KG(z) = 1 + \frac{(0.368z + 0.264)K}{z^2 - 1.368z + 0.368} = 0$$

or

$$z^2 + (0.368K - 1.368)z + (0.368 + 0.264K) = 0$$

The Jury array is

z^0	z^1	z^2
$0.368 + 0.264K$	$0.368K - 1.368$	1

2nd-order → 1 row and 3 constraints.

The constraint $Q(1) > 0$ yields

$$1 + (0.368K - 1.368) + (0.368 + 0.264K) = 0.632K > 0 \Rightarrow K > 0$$

The constraint $(-1)^2 Q(-1) > 0$ yields

$$1 - 0.368K + 1.368 + 0.368 + 0.264K > 0 \Rightarrow K < \frac{2.736}{0.104} = 26.3$$

The constraint $|a_0| < a_2$ yields

$$0.368 + 0.264K < 1 \Rightarrow K < \frac{0.632}{0.264} = 2.39$$

Thus the system is stable for

$$0 < K < 2.39$$

The system is marginally stable for $K = 2.39$. For this value of K , the characteristic equation is

$$z^2 + (0.368K - 1.368)z + (0.368 + 0.264K)|_{K=2.39} = z^2 - 0.488z + 1 = 0$$

The roots of this equation are

On the unit circle

$$z = 0.244 \pm j0.970 = 1/\pm 75.9^\circ = 1/\pm 1.32 \text{ rad} = 1/\pm \omega T$$

Since $T = 1$ s, the system will oscillate at a frequency of 1.32 rad/s. These results verify those of Example 7.3.

Example 7.5

Suppose that the characteristic equation for a closed-loop discrete-time system is given by the expression

$$Q(z) = z^3 - 1.8z^2 + 1.05z - 0.20 = 0$$

The first conditions of Jury's test are

$$\left\{ \begin{array}{l} Q(1) = 1 - 1.8 + 1.05 - 0.2 = 0.05 > 0 \\ (-1)^3 Q(-1) = -[-1 - 1.8 - 1.05 - 0.2] = 4.05 > 0 \\ |a_0| = 0.2 < a_3 = 1 \end{array} \right.$$

The Jury array is calculated to be

z^0	z^1	z^2	z^3
-0.2	1.05	-1.8	1
1	-1.8	1.05	-0.2
-0.96	1.59	-0.69	

where the last row has been calculated as follows:

$$b_0 = \begin{vmatrix} -0.2 & 1 \\ 1 & -0.2 \end{vmatrix} = -0.96, \quad b_1 = \begin{vmatrix} -0.2 & -1.8 \\ 1 & 1.05 \end{vmatrix} = 1.59$$

$$b_2 = \begin{vmatrix} -0.2 & 1.05 \\ 1 & -1.8 \end{vmatrix} = -0.69$$

Hence the last condition is

$$\underline{|b_0| = 0.96 > |b_2| = 0.69}$$

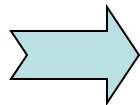
Therefore, since all conditions are satisfied, the system is stable. The characteristic equation can be factored as

$$\underline{Q(z) = (z - 0.5)^2(z - 0.8)}$$

This form of the equation clearly indicates the system's stability.

CT (S-plane): Routh-Hurwitz Criterion

DT (Z-plane): Jury's Test



For high-order systems

7.6 Root Locus

For the LTI sampled-data system of figure 7-6

$$\frac{C(z)}{R(z)} = \frac{KG(z)}{1 + K\overline{GH}(z)}$$

The system characteristic equation is

$$1 + K\overline{GH}(z) = 0 \quad (7.16)$$

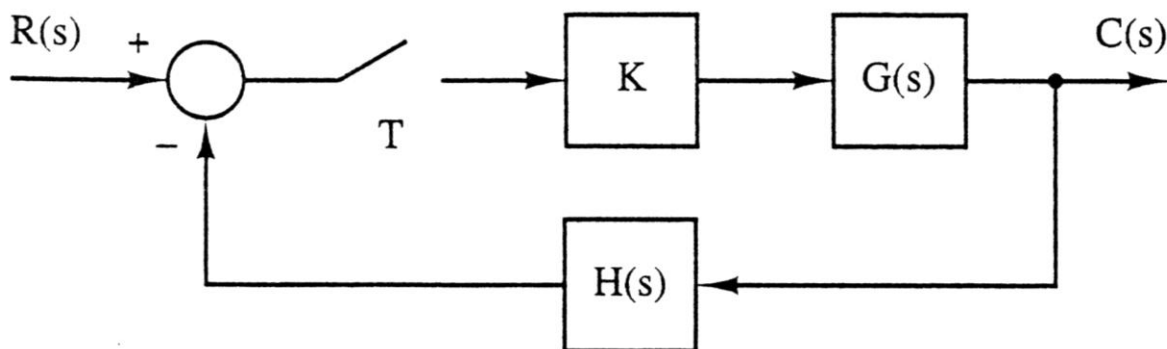


Figure 7-6 Sampled-data system.

TABLE 7-3 RULES FOR ROOT-LOCUS CONSTRUCTION

For the characteristic equation

$$1 + K\overline{GH}(z) = 0$$

1. Loci originate on poles of $\overline{GH}(z)$ and terminate on the zeros of $\overline{GH}(z)$.
2. The root locus on the real axis always lies in a section of the real axis to the left of an odd number of poles and zeros on the real axis.
3. The root locus is symmetrical with respect to the real axis.
4. The number of asymptotes is equal to the number of poles of $\overline{GH}(z)$, n_p , minus the number of zeros of $\overline{GH}(z)$, n_z , with angles given by $(2k + 1)\pi/(n_p - n_z)$.
5. The asymptotes intersect the real axis at σ , where

$$\sigma = \frac{\sum \text{poles of } \overline{GH}(z) - \sum \text{zeros of } \overline{GH}(z)}{n_p - n_z}$$

6. The breakaway points are given by the roots of

$$\frac{d[\overline{GH}(z)]}{dz} = 0$$

or, equivalently,

$$D(z) \frac{dN(z)}{dz} - N(z) \frac{dD(z)}{dz} = 0, \quad \overline{GH}(z) = \frac{N(z)}{D(z)}$$

8.1 INTRODUCTION

- THE CONTROL SYSTEM PROBLEM

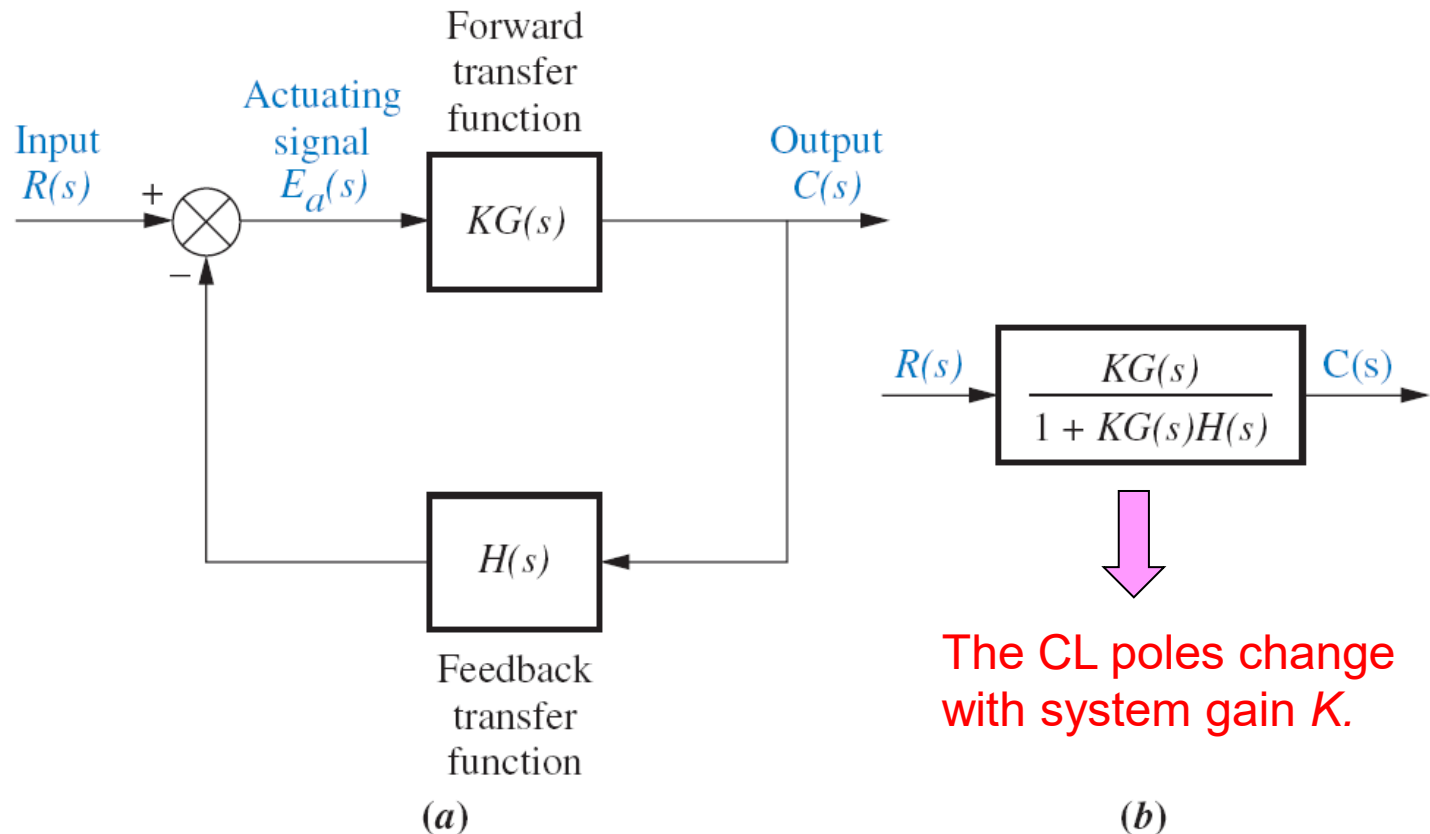


FIGURE 8.1 a. Closed-loop system; b. equivalent transfer function

N: Numerator

D: Denominator

$$G(s) = \frac{N_G(s)}{D_G(s)} \quad (8.1)$$

$$H(s) = \frac{N_H(s)}{D_H(s)} \quad (8.2)$$

↗ The CL zeros = G's zeros + H's poles.

$$T(s) = \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + KN_G(s)N_H(s)} \quad (8.3)$$

↘ The CL poles change with system gain K .

8.2 DEFINING THE ROOT LOCUS

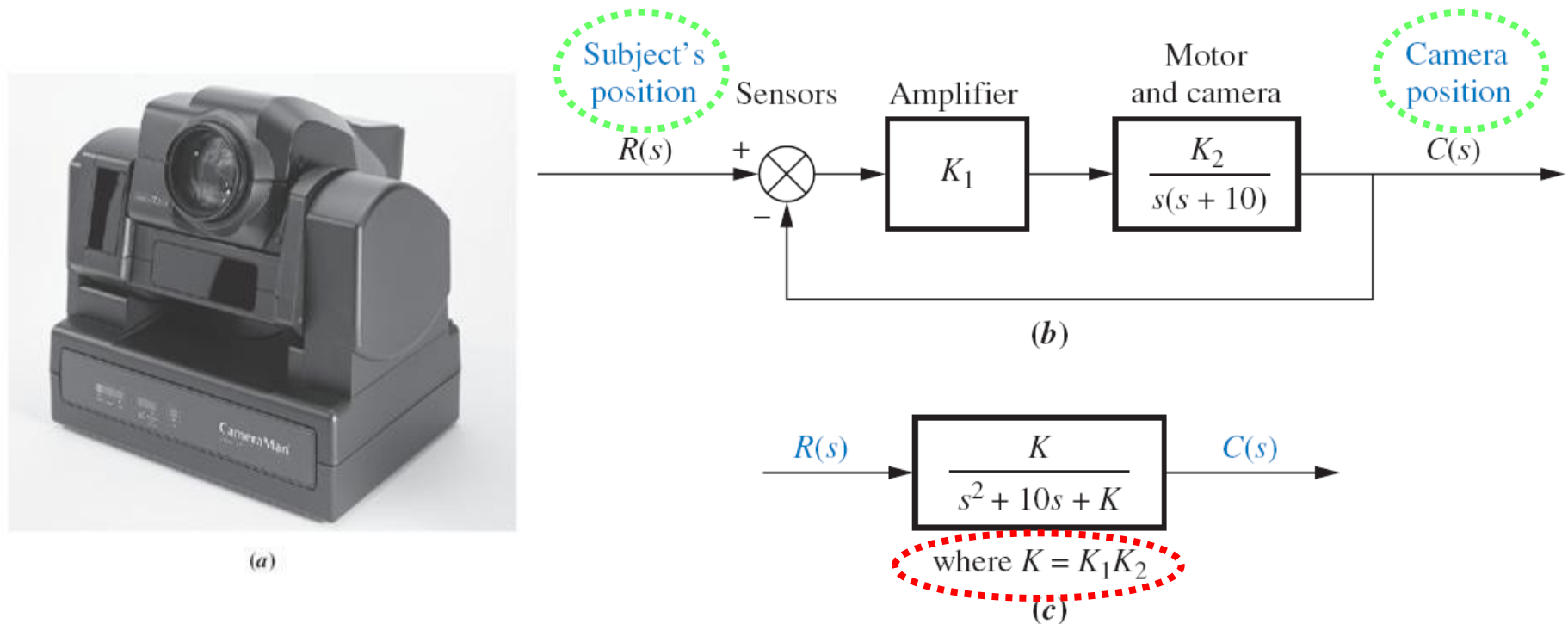


FIGURE 8.4 **a.** The CameraMan[®] Presenter Camera System automatically follows a subject who wears infrared sensors on the front and back (the front sensor is also a microphone); tracking commands and audio are relayed to CameraMan via a radio frequency link from a unit worn by the subject; **b.** block diagram; **c.** closed-loop transfer function

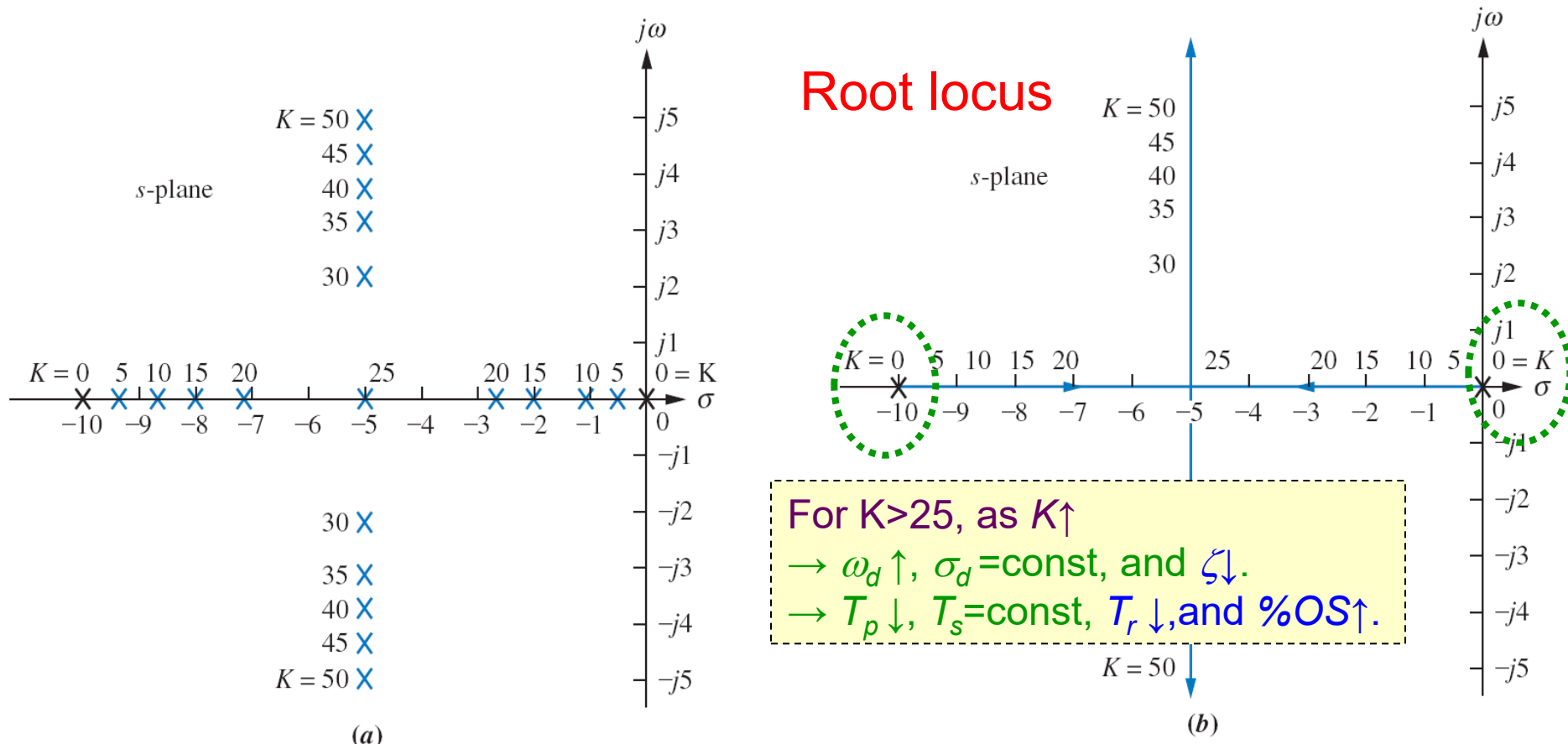
TABLE 8.1 Pole location as function of gain for the system of Figure 8.4

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

$K = 0$,
open-loop
poles

For $K > 0$

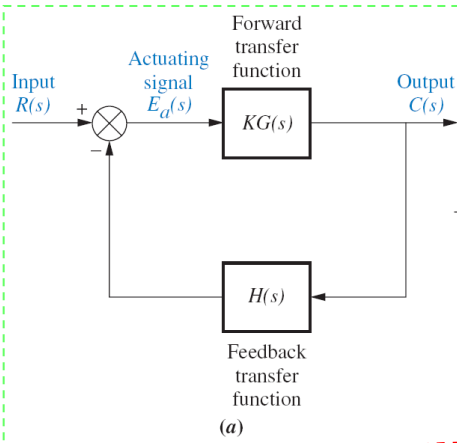
FIGURE 8.5 **a.** Pole plot from Table 8.1; **b.** root locus



Root locus: the paths of the CL poles as the gain is varied.

Review Example 5.4.

8.3 PROPERTIES OF THE ROOT LOCUS



$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)} \quad (8.12)$$

KGH : Open-loop TF (loop gain)

$$KG(s)H(s) = -1 = 1 \angle (2k + 1)180^\circ \quad k = 0, \pm 1, \pm 2, \pm 3, \dots \quad (8.13)$$

For closed-loop poles



$$|KG(s)H(s)| = 1 \quad (8.14)$$

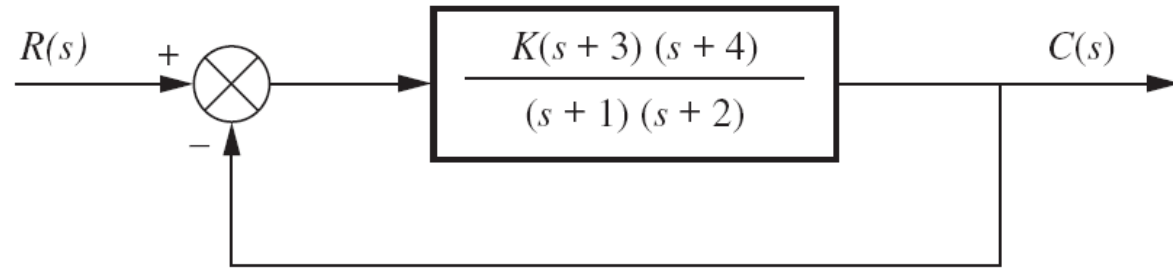
$$\angle KG(s)H(s) = (2k + 1)180^\circ \quad (8.15)$$

$$K = \frac{1}{|G(s)||H(s)|} \quad (8.16)$$

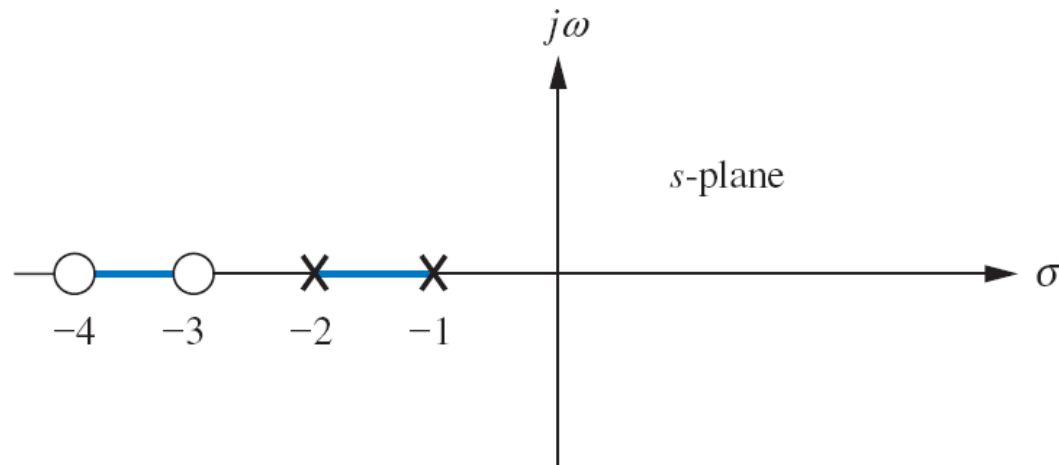
Example: *CameraMan*

$$KG(s)H(s) = \frac{K}{s(s + 10)} \quad (8.17)$$

An Example



(a)



(b)

FIGURE 8.6 a. Example system; b. pole-zero plot of $G(s)$

Open-loop TF

$$KG(s)H(s) = \frac{K(s+3)(s+4)}{(s+1)(s+2)} \quad (8.18)$$

Closed-loop TF

$$T(s) = \frac{K(s+3)(s+4)}{(1+K)s^2 + (3+7K)s + (2+12K)} \quad (8.19)$$

Consider the point: $-2 + j3$

$$\theta_1 + \theta_2 - \theta_3 - \theta_4 = 56.31^\circ + 71.57^\circ - 90^\circ - 108.43^\circ = -70.55^\circ \quad (8.20)$$

➡ Not a root locus point (not a closed-loop pole for any gain K).

$$K = \frac{1}{|G(s)H(s)|} = \frac{1}{M} = \frac{\prod \text{pole lengths}}{\prod \text{zero lengths}} \quad (8.21)$$

Case1:

$$-2 + j3$$

➡ Not a root locus point.

Case2:

$$-2 + j(\sqrt{2}/2)$$

➡ A root locus point.
 $K = 0.33$

- Summary

- 1) Check angle

- 2) Find $K = 1 / |GH|$

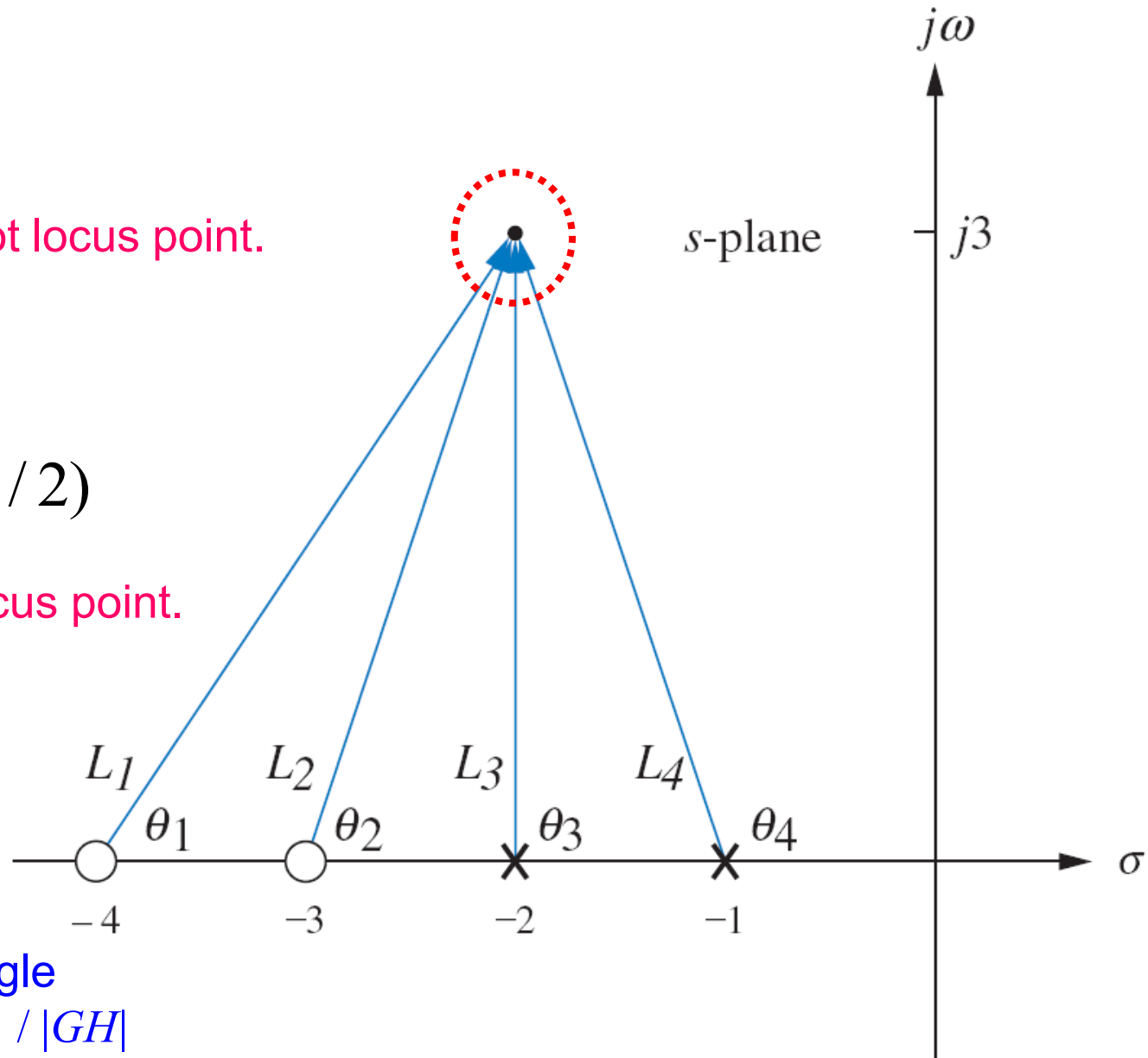


FIGURE 8.7 Vector representation of $G(s)$ from Figure 8.6(a) at $-2 + j3$

8.4 SKETCHING THE ROOT LOCUS

- *Sketch* the root locus w/o the effort to *plot* the locus.

- **Five Rules**

1) Num. of branches: *num. of closed-loop poles.*

2) Symmetry: *about the real axis.*

Physically realizable systems
cannot have complex coefficients.

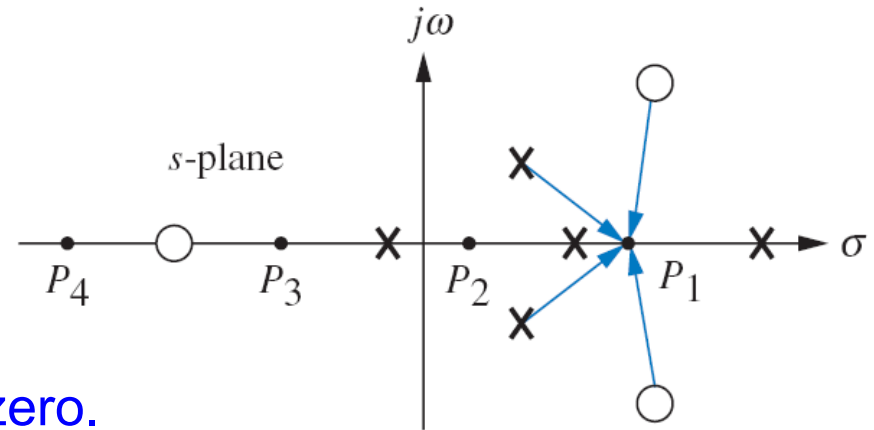
3) Real-axis segments: *left of an odd num. of poles/zeros. [see Fig. 8.8-9]*

4) Starting & ending points: *begins at the GH poles and ends at the GH zeros, where GH is the open-loop TF. [see Fig. 8.10]*

5) Behavior at infinity: *approaches straight lines as asymptotes.*

3) Real-axis Segments

FIGURE 8.8 Poles and zeros of a general open-loop system with test points, P_i , on the real axis



- Observations

1) Angle of *conjugate* poles/zeros is zero.

2) Angle of poles/zeros *at the left* of the test (respective) points is zero.

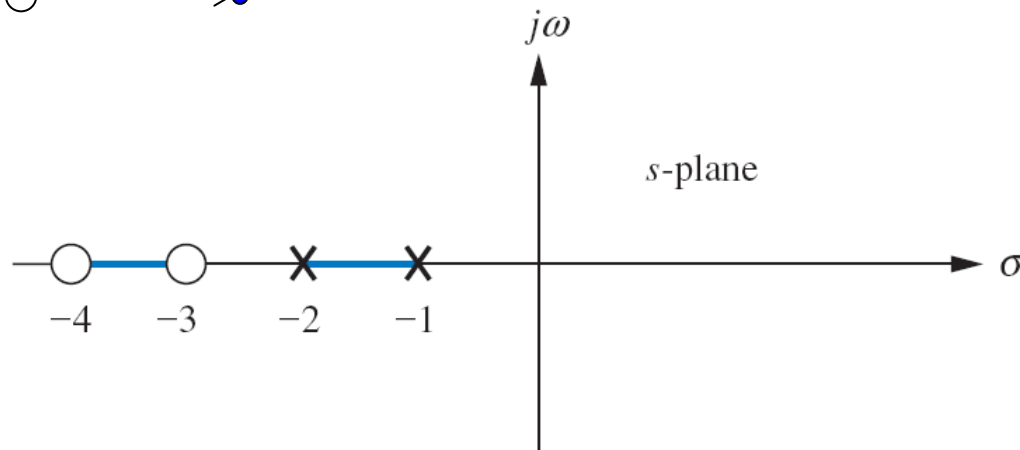
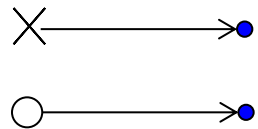


FIGURE 8.9 Real-axis segments of the root locus for the system of Figure 8.6

4) Starting and Ending Points

$$T(s) = \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + KN_G(s)N_H(s)} \quad (8.3)$$

$\times$
 $\downarrow$
 $\circ$

• K is small

• K is large

$$T(s) \approx \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + \epsilon} \quad (8.23)$$

$$T(s) \approx \frac{KN_G(s)D_H(s)}{\epsilon + KN_G(s)N_H(s)} \quad (8.24)$$

poles

zeros

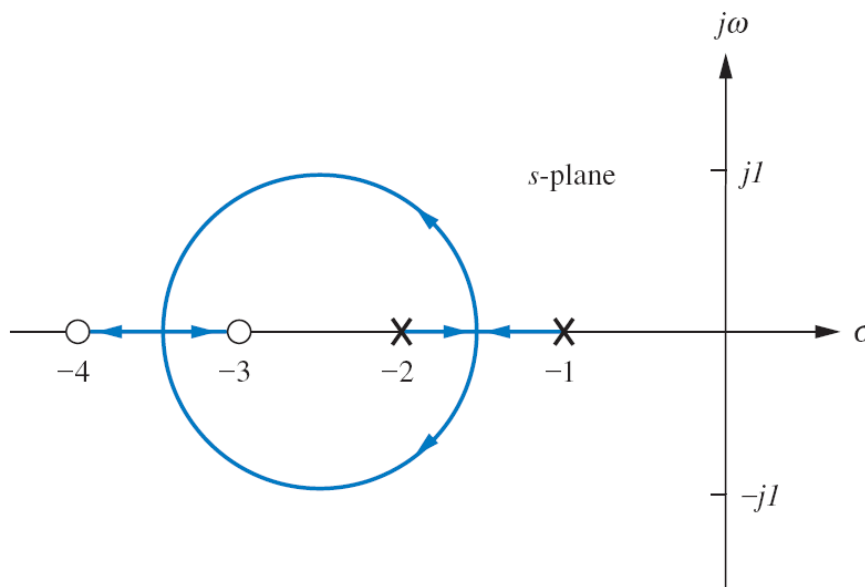


FIGURE 8.10 Complete root locus for the system of Figure 8.6

5) Behavior at Infinity

As $s \rightarrow \infty$,

if $GH \rightarrow \infty$, $s = \text{pole}$,

if $GH \rightarrow 0$, $s = \text{zero}$. ★

$$KG(s)H(s) = \frac{K}{s(s+1)(s+2)} \quad (8.25)$$

$$KG(s)H(s) \approx \frac{K}{s^3} = \frac{K}{s \cdot s \cdot s} \quad (8.26)$$

----- Physical systems have more finite poles than zeros.

Asymptotes

$$\sigma_a = \frac{\Sigma \text{ finite poles} - \Sigma \text{ finite zeros}}{\# \text{ finite poles} - \# \text{ finite zeros}} \quad (8.27)$$

$$\theta_a = \frac{(2k+1)\pi}{\# \text{ finite poles} - \# \text{ finite zeros}} \quad (8.28)$$

where $k = 0, \pm 1, \pm 2, \pm 3$ and the angle is given in radians with respect to the positive extension of the real axis.

Details of derivation: See Appendix L.1.

EXAMPLE 8.2

Sketching a root locus with asymptotes

Problem: Sketch the root locus for the system shown in Figure 8.11.

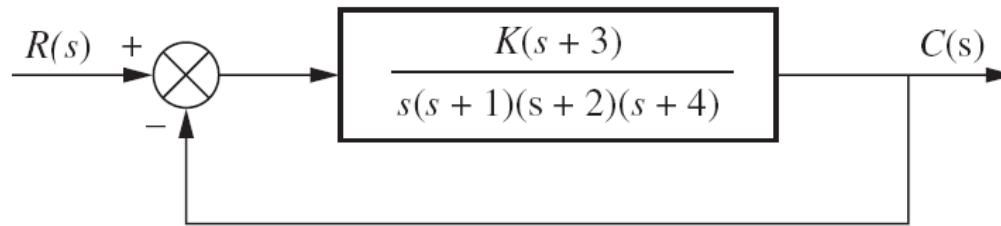


Figure 8.11 System for Example 8.2.

SOLUTION: Let us begin by calculating the asymptotes. Using Eq. (8.27), the real-axis intercept is evaluated as

$$\sigma_a = \frac{(-1 - 2 - 4) - (-3)}{4 - 1} = -\frac{4}{3} \quad (8.29)$$

EXAMPLE 8.2

The angles of the lines that intersect at $-4/3$, given by Eq. (8.28), are

$$\theta_a = \frac{(2k + 1)\pi}{\# \text{finite poles} - \# \text{finite zeros}} \quad (8.30a)$$

$$= \pi/3 \quad \text{for } k = 0 \quad (8.30b)$$

$$= \pi \quad \text{for } k = 1 \quad (8.30c)$$

$$= 5\pi/3 \quad \text{for } k = 2 \quad (8.30d)$$

asymptotes = # finite poles – # finite zeros

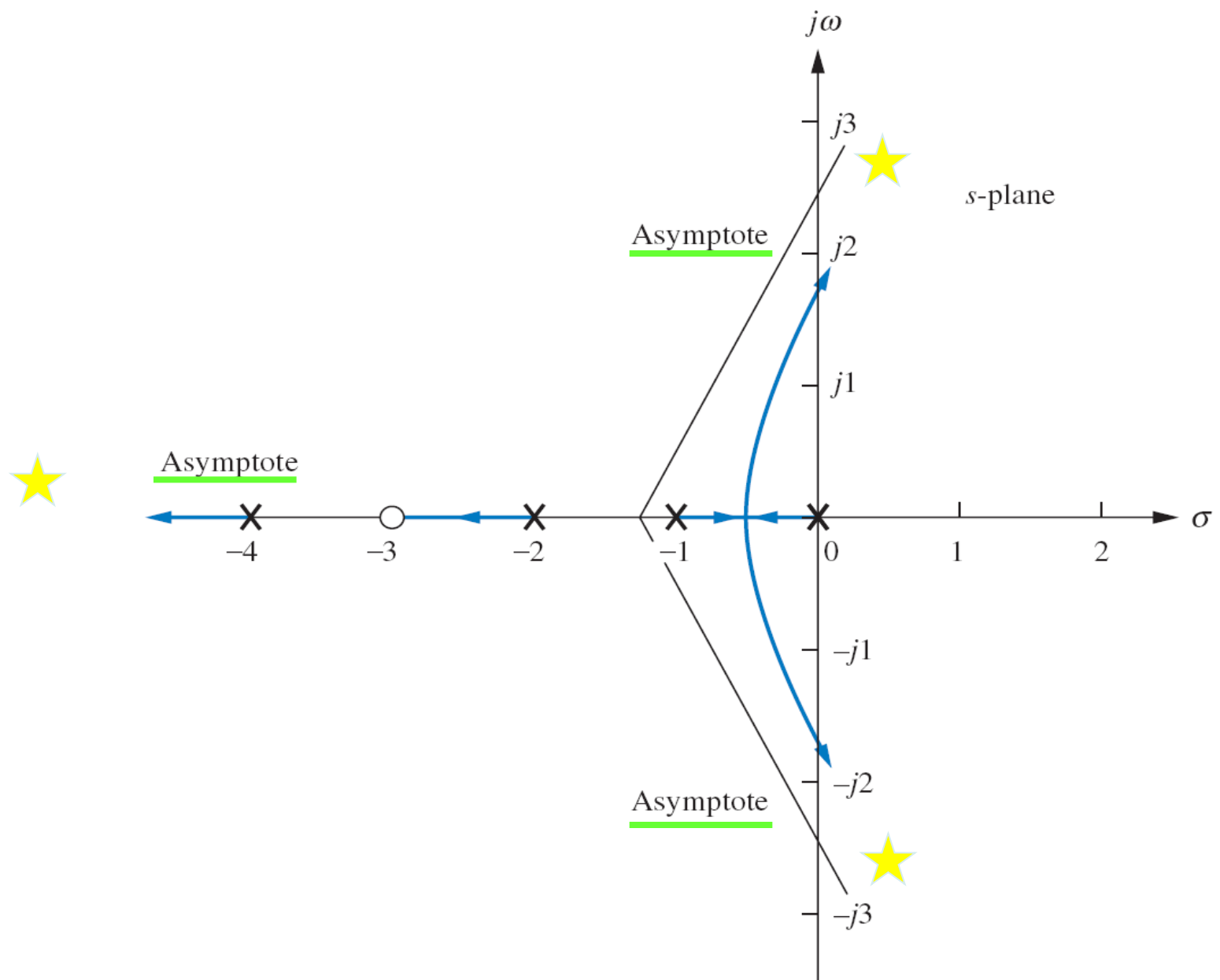


FIGURE 8.12 Root locus and asymptotes for the system of Figure 8.11

8.5 REFINING THE SKETCH

1) REAL-AXIS BREAKAWAY AND BREAK-IN POINTS

Angle: $180^\circ / n$,
where n is the num. of CL poles
arriving at or departing from the
single breakaway or break-in point
on the real axis.

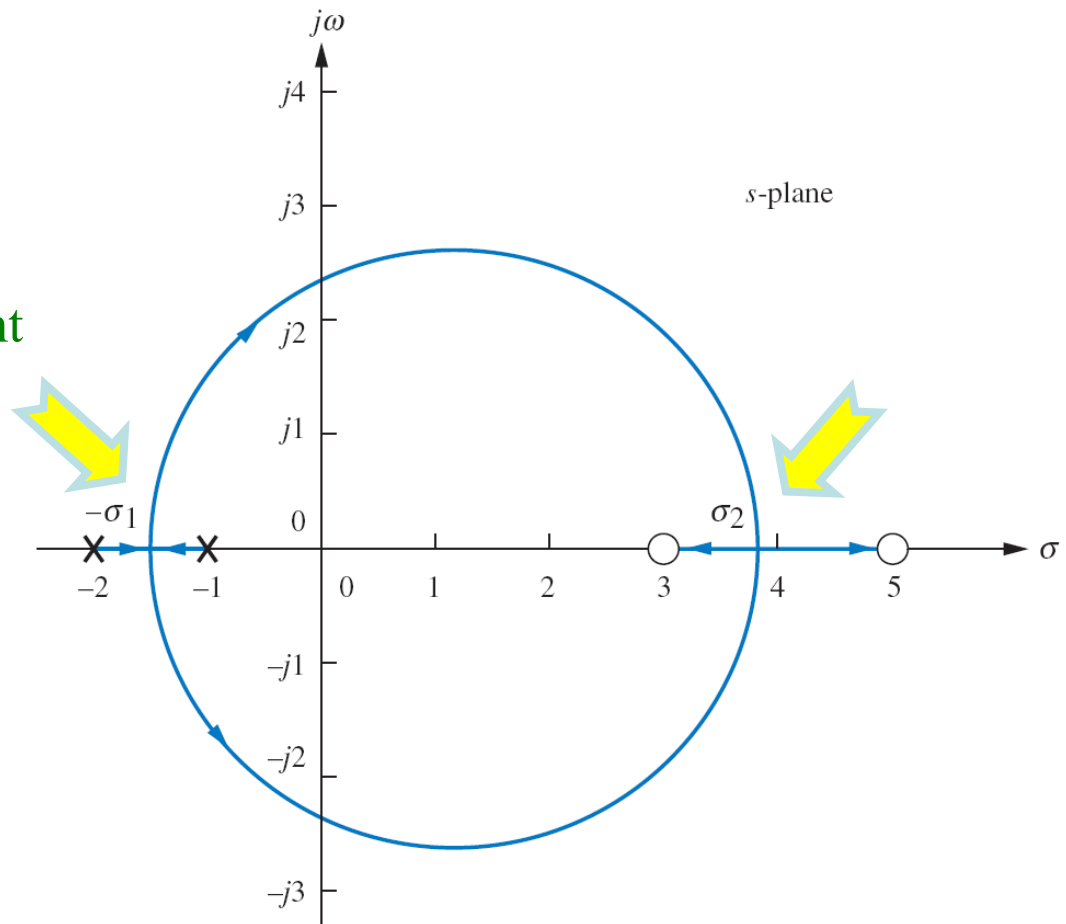
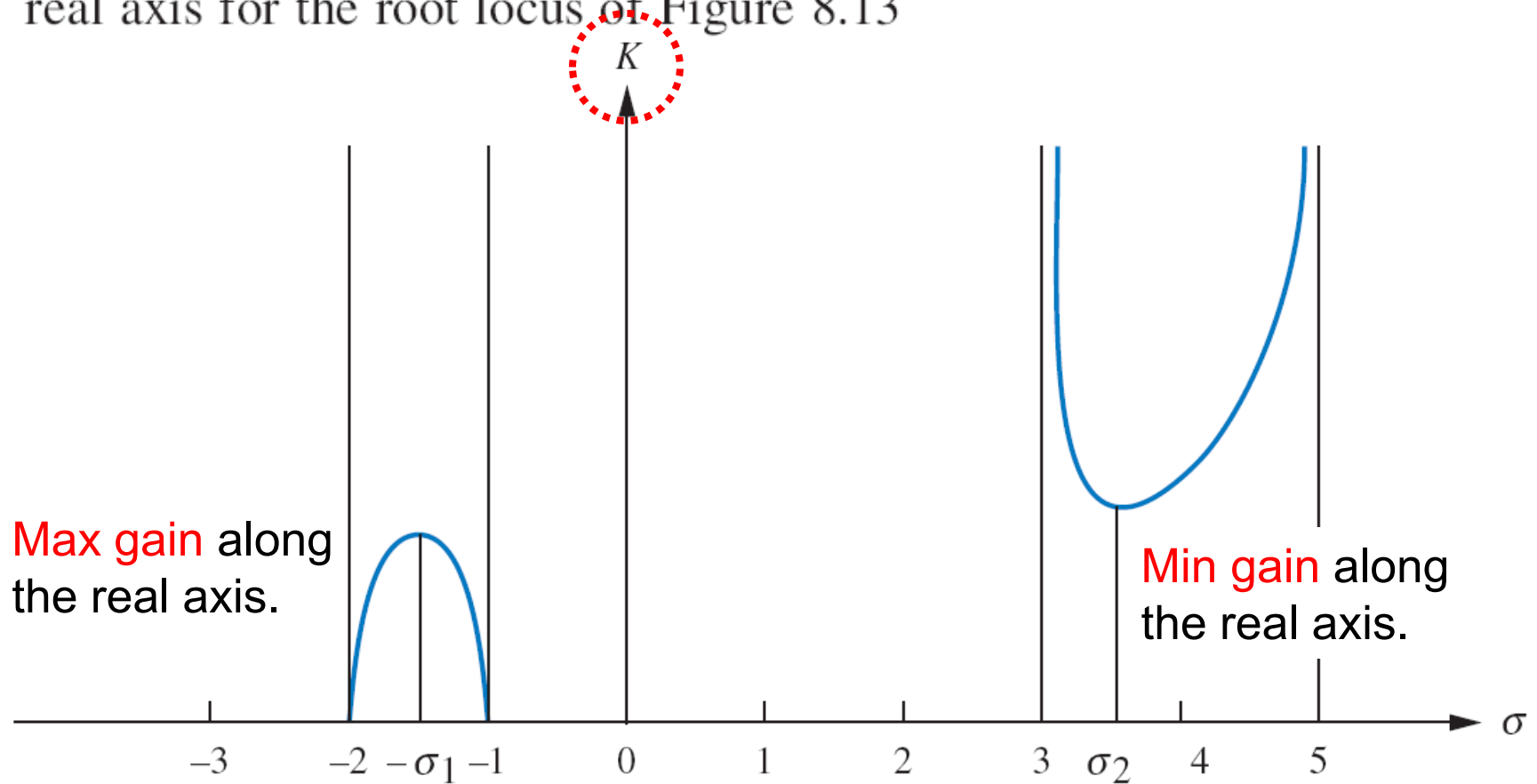


FIGURE 8.13 Root locus example showing real-axis breakaway ($-\sigma_1$) and break-in points (σ_2)

FIGURE 8.14 Variation of gain along the real axis for the root locus of Figure 8.13



$$K = -\frac{1}{G(s)H(s)} \quad (8.31)$$

$$K = -\frac{1}{G(\sigma)H(\sigma)} \quad (8.32)$$



$$\frac{dK}{d\sigma} = 0$$

EXAMPLE 8.3

Breakaway and break-in points via differentiation

Problem: Find the breakaway and break-in points for the root locus of Figure 8.13, using differential calculus.

SOLUTION: Using the open-loop poles and zeros, we represent the open-loop system whose root locus is shown in Figure 8.13 as follows:

$$KG(s)H(s) = \frac{K(s-3)(s-5)}{(s+1)(s+2)} = \frac{K(s^2 - 8s + 15)}{(s^2 + 3s + 2)} \quad (8.33)$$

But for all points along the root locus, $KG(s)H(s) = -1$, and along the real axis, $s = \sigma$. Hence,

$$\frac{K(\sigma^2 - 8\sigma + 15)}{(\sigma^2 + 3\sigma + 2)} = -1 \quad (8.34)$$

EXAMPLE 8.3

Solving for K , we find

$$K = \frac{-(\sigma^2 + 3\sigma + 2)}{(\sigma^2 - 8\sigma + 15)} \quad (8.35)$$

Differentiating K with respect to σ and setting the derivative equal to zero yields

$$\frac{dK}{d\sigma} = \frac{(11\sigma^2 - 26\sigma - 61)}{(\sigma^2 - 8\sigma + 15)^2} = 0 \quad (8.36)$$

Solving for σ , we find $\sigma = -1.45$ and 3.82 , which are the breakaway and break-in points.

EXAMPLE 8.5

2) THE $j\omega$ -AXIS CROSSINGS

Continue Example 8.2

Frequency and gain at imaginary-axis crossing

Problem: For the system of Figure 8.11, find the frequency and gain, K , for which the root locus crosses the imaginary axis. For what range of K is the system stable?

SOLUTION: The closed-loop transfer function for the system of Figure 8.11 is

$$T(s) = \frac{K(s + 3)}{s^4 + 7s^3 + 14s^2 + (8 + K)s + 3K} \quad (8.40)$$

Using the denominator and simplifying some of the entries by multiplying any row by a constant, we obtain the Routh array shown in Table 8.3.

A complete row of zeros yields the possibility for imaginary axis roots. For positive values of gain, those for which the root locus is plotted, only the s^1 row can yield a row of zeros. Thus,

$$-K^2 - 65K + 720 = 0 \quad (8.41)$$

EXAMPLE 8.5

TABLE 8.3 Routh table for Eq. (8.40)

s^4	1	14	$3K$
s^3	7	$8 + K$	
s^2	$90 - K$	$21K$	
s^1	$\frac{-K^2 - 65K + 720}{90 - K}$		
s^0	$21K$		

EXAMPLE 8.5

From this equation K is evaluated as

$$K = 9.65 \quad (8.42)$$

Forming the even polynomial by using the s^2 row with $K = 9.65$, we obtain

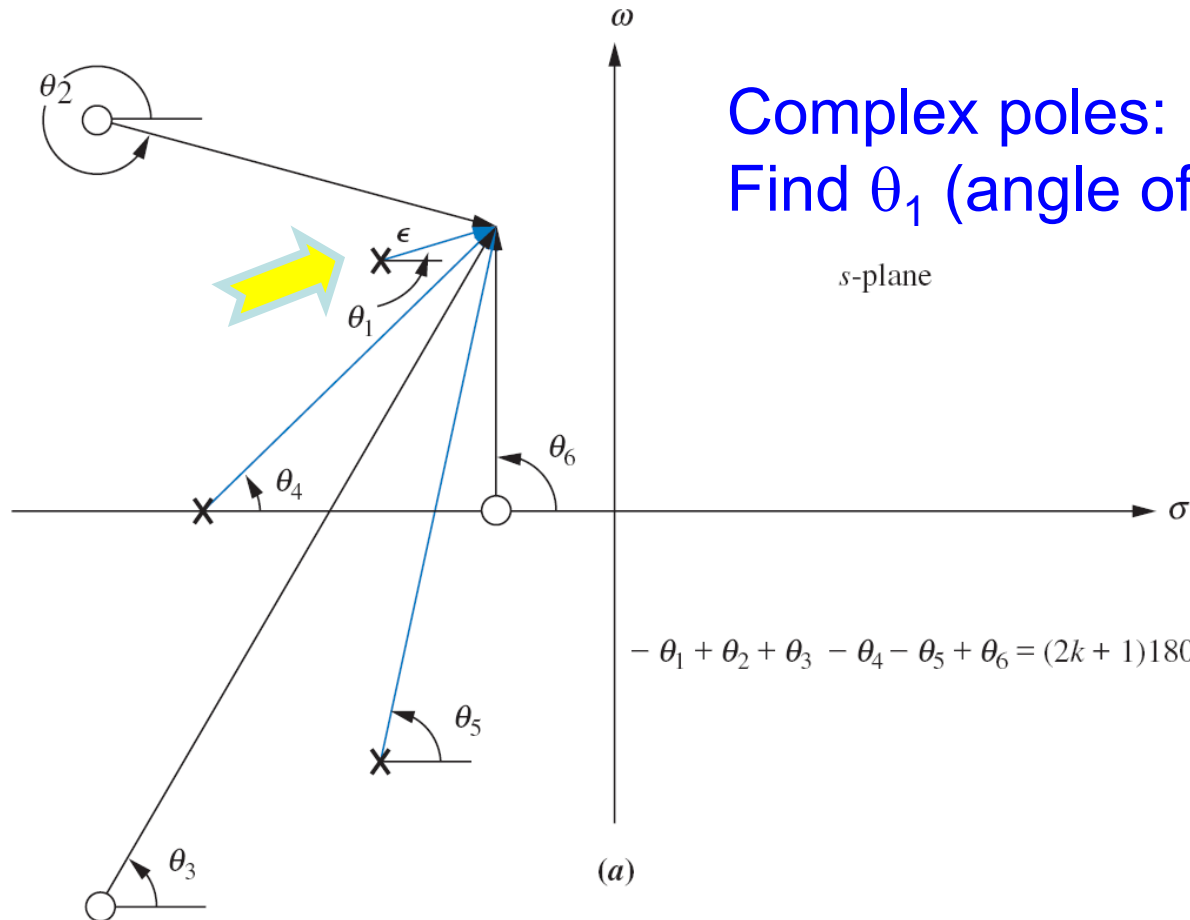
$$(90 - K)s^2 + 21K = 80.35s^2 + 202.7 = 0 \quad (8.43)$$

and s is found to be equal to $\pm j1.59$. Thus the root locus crosses the $j\omega$ -axis at $\pm j1.59$ at a gain of 9.65. We conclude that the system is stable for $0 \leq K < 9.65$.

3) ANGLES OF DEPARTURE AND ARRIVAL

$$-\theta_1 + \theta_2 + \theta_3 - \theta_4 - \theta_5 + \theta_6 = (2k + 1)180^\circ \quad (8.44a)$$

$$\theta_1 = \theta_2 + \theta_3 - \theta_4 - \theta_5 + \theta_6 - (2k + 1)180^\circ \quad (8.44b)$$



Complex poles:
Find θ_1 (angle of departure)

s-plane

$$-\theta_1 + \theta_2 + \theta_3 - \theta_4 - \theta_5 + \theta_6 = (2k + 1)180$$

(a)

$$-\theta_1 + \theta_2 + \theta_3 - \theta_4 - \theta_5 + \theta_6 = (2k + 1)180^\circ \quad (8.45a)$$

$$\theta_2 = \theta_1 - \theta_3 + \theta_4 + \theta_5 - \theta_6 + (2k + 1)180^\circ \quad (8.45b)$$

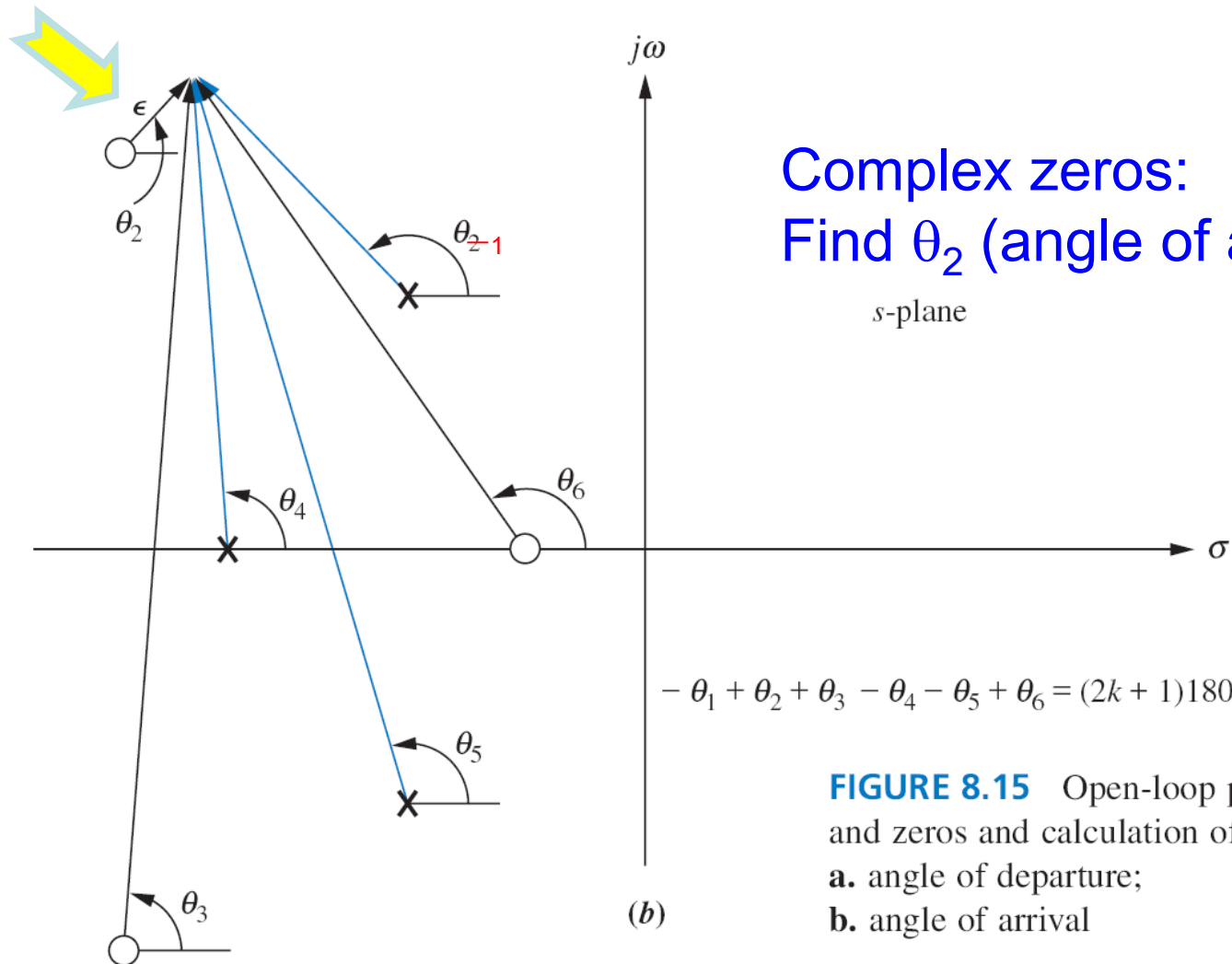


FIGURE 8.15 Open-loop poles and zeros and calculation of
a. angle of departure;
b. angle of arrival

Example 8.6

Angle of departure from a complex pole

Problem: Given the unity feedback system of Figure 8.16, find the angle of departure from the complex poles and sketch the root locus.

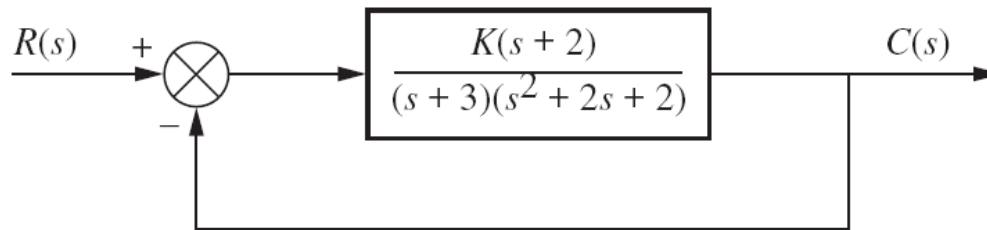
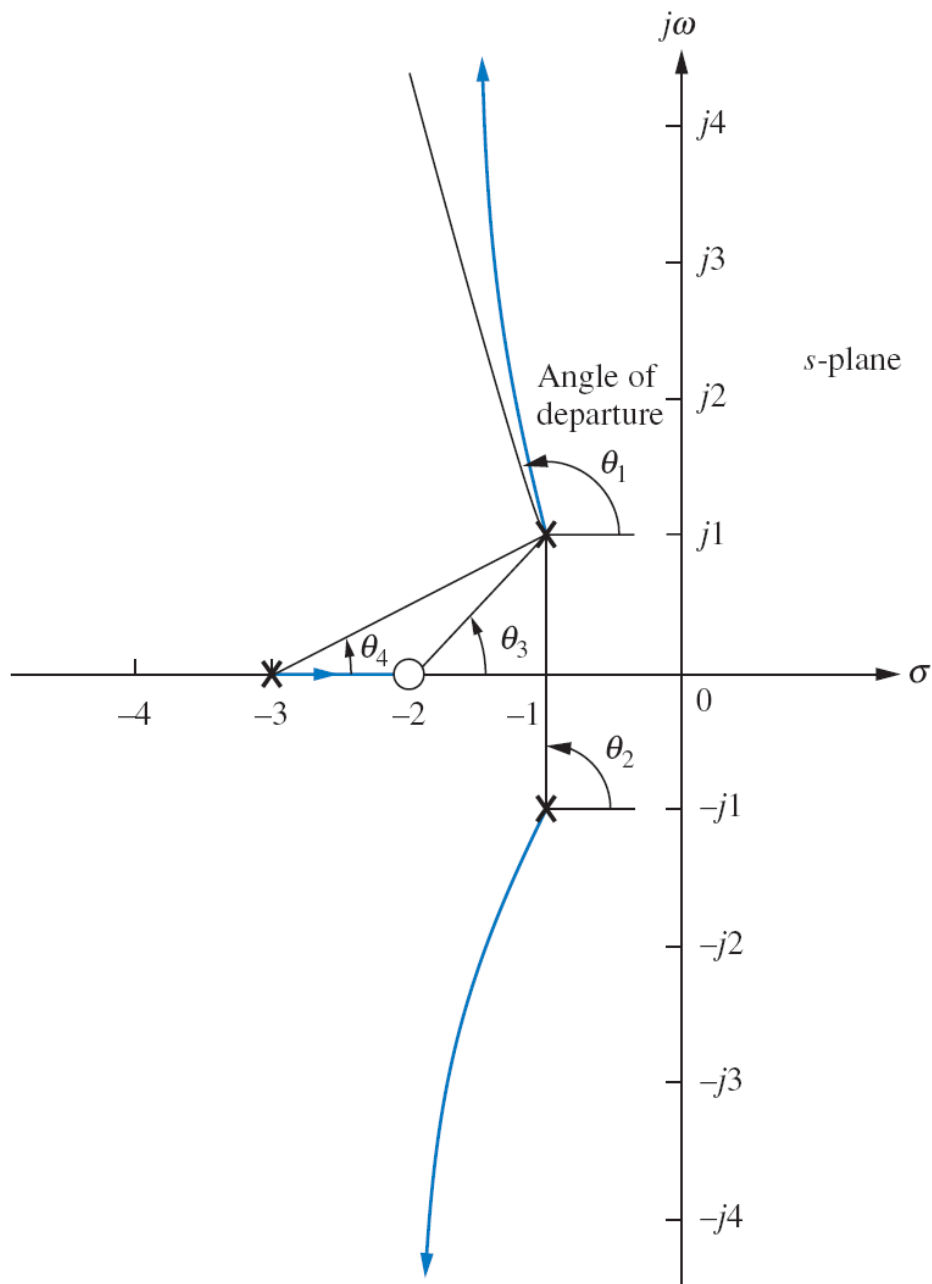


FIGURE 8.16 Unity feedback system with complex poles

SOLUTION: Using the poles and zeros of $G(s) = (s + 2)/[(s + 3)(s^2 + 2s + 2)]$ as plotted in Figure 8.17, we calculate the sum of angles drawn to a point ϵ close to the complex pole, $-1 + j1$, in the second quadrant. Thus,

$$-\theta_1 - \theta_2 + \theta_3 - \theta_4 = -\theta_1 - 90^\circ + \tan^{-1}\left(\frac{1}{1}\right) - \tan^{-1}\left(\frac{1}{2}\right) = 180^\circ \quad (8.46)$$

from which $\theta_1 = -251.6^\circ = 108.4^\circ$. A sketch of the root locus is shown in Figure 8.17. Notice how the departure angle from the complex poles helps us to refine the shape.



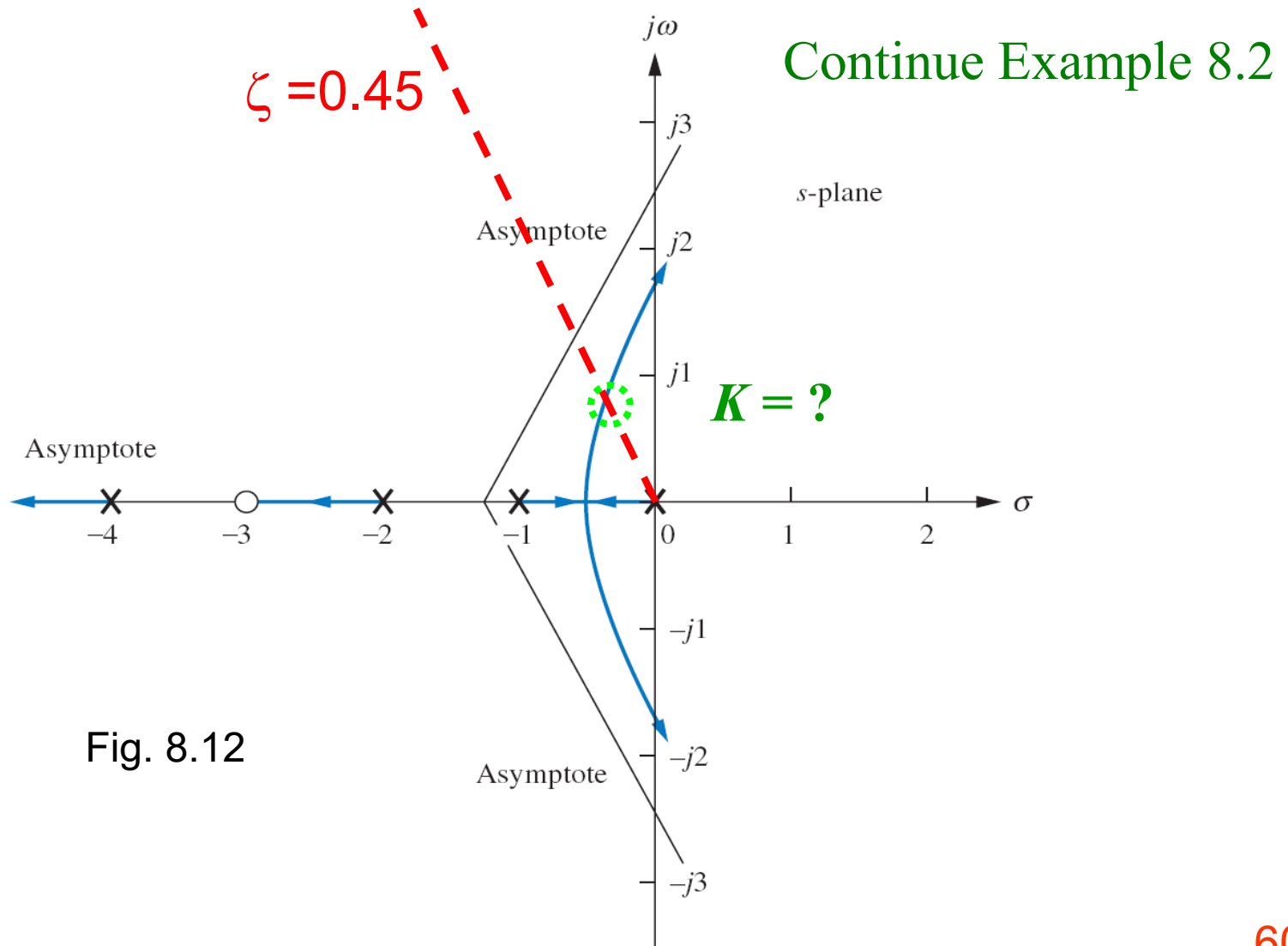
Asymptotes:

$$\sigma_a = -3/2,$$

$$\theta_a = \pi/2, 3\pi/2$$

FIGURE 8.17 Root locus for system of Figure 8.16 showing angle of departure

4) PLOTTING AND CALIBRATING THE ROOT LOCUS



$$r=2, \quad \theta_2 - \theta_1 - \theta_3 - \theta_4 - \theta_5 = -251.5^\circ \quad (8.47)$$

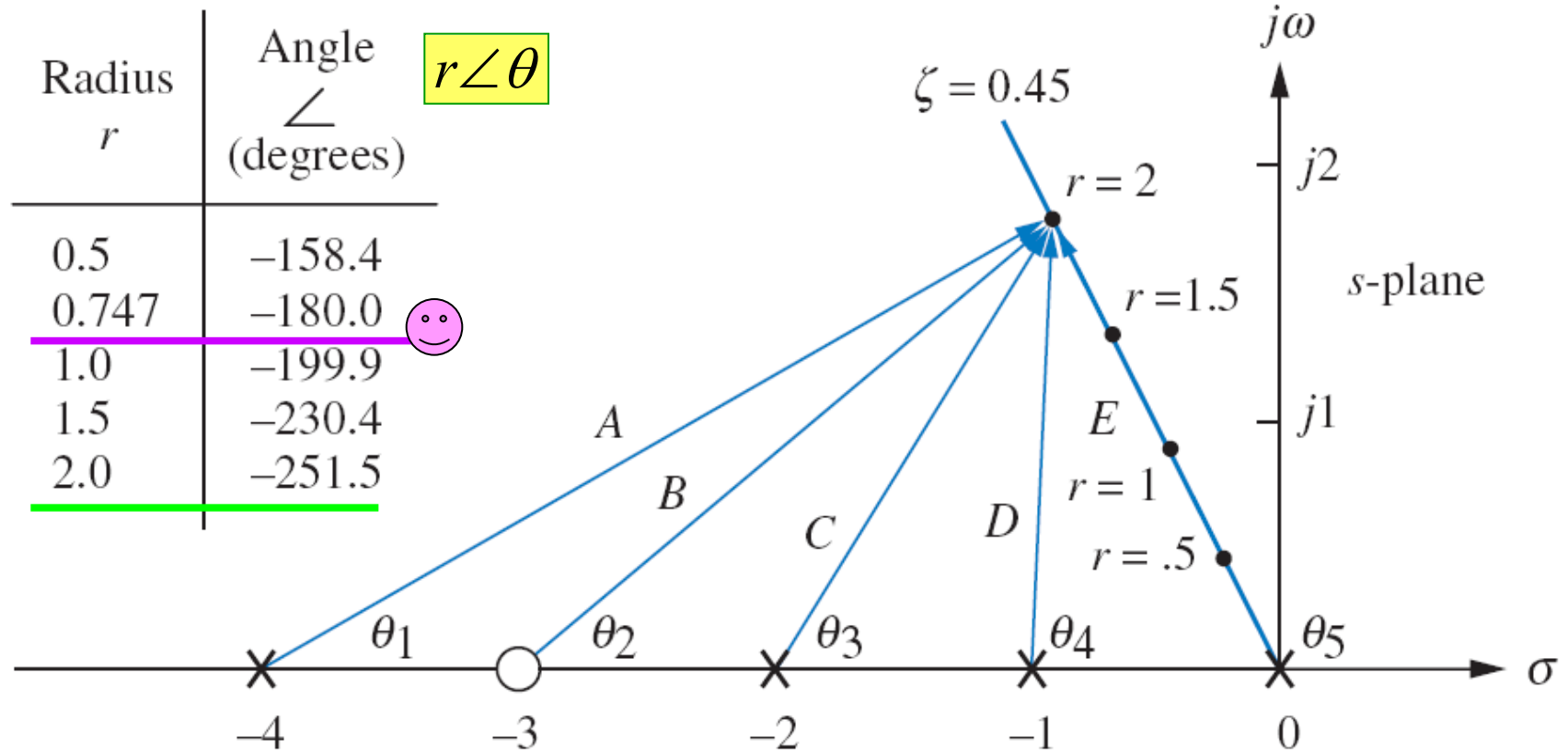


FIGURE 8.18 Finding and calibrating exact points on the root locus of Figure 8.12

$$K = \frac{|A||C||D||E|}{|B|} = 1.71 \quad (8.48)$$

Example 7.7

Consider again the system of Example 6.4. For this system

$$KG(z) = \frac{0.368K(z + 0.717)}{(z - 1)(z - 0.368)}$$

Thus the loci originate at $z = 1$ and $z = 0.368$, and terminate at $z = -0.717$ and $z = \infty$. There is one asymptote, at 180° . The breakaway points, obtained from

$$\frac{d}{dz}[G(z)] = 0$$

Marginally Stable

Method-1

occur at $z = 0.65$ for $K = 0.196$, and $z = -2.08$ for $K = 15.0$. The root locus is then as shown in Figure 7-7. The points of intersection of the root loci with the unit circle may be found by graphical construction, the Jury stability test, or the Routh–Hurwitz criterion. To illustrate the use of the Jury stability test, consider the results of Example 7.4. The value of gain for marginal stability (i.e., for the roots to appear on the unit circle) is $K = 2.39$. For this value of gain, the characteristic equation is, from Example 7.4,

$$z^2 - 0.488z + 1 = 0$$

The roots of this equation are

$$z = 0.244 \pm j0.970 = 1 \angle \pm 75.8^\circ = 1 \angle \pm 1.32 \text{ rad} = 1 \angle \pm \omega T$$

and thus these are the points at which the root locus crosses the unit circle. Note that the frequency of oscillation for this case is $\omega = 1.32$, since $T = 1$ s. This was also calculated in Example 7.3 using the Routh–Hurwitz criterion, and in Example 7.4 using the Jury test.

Method-2

The value of the gain at points where the root locus crosses the unit circle can also be determined using the root-locus condition that at any point along the locus the magnitude of the open-loop function must be equal to 1 [i.e., $|K\overline{GH}(z)| = 1$]. Using the condition and Figure 7-8, we note that

$$\frac{0.368K(Z_1)}{(P_1)(P_2)} = 1$$

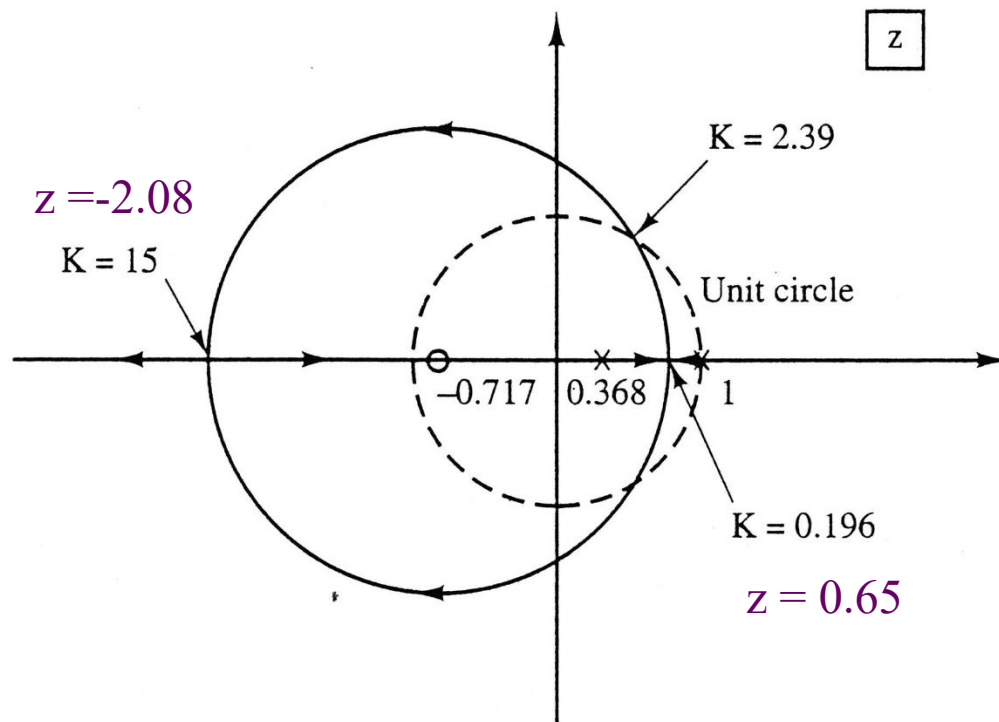


Figure 7-7 Root locus for Example 7.7.

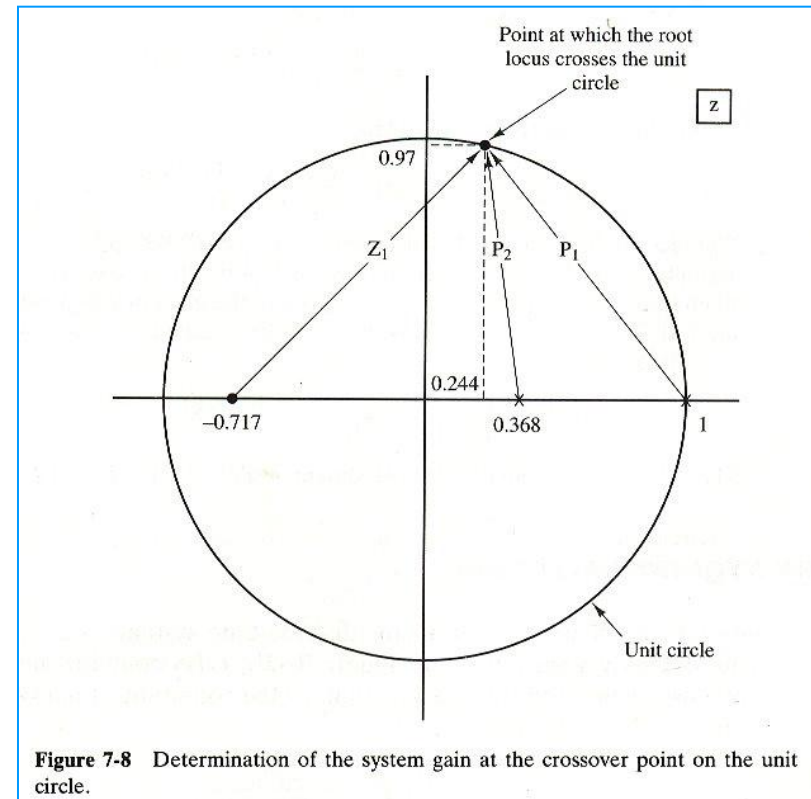


Figure 7-8 Determination of the system gain at the crossover point on the unit circle.

Example 7.8

Consider the system shown in Figure 7-9. The transfer function $[1 + 10s]$ is that of an analog proportional-plus-derivative (PD) controller [1]. The digital PD controller is covered in Chapter 8. We wish to determine the form of the root locus and the range of K for stability.

We denote the open-loop function as

$$KG(s) = \frac{1 - e^{-sT}}{s} \left[\frac{K(1 + 10s)}{s^2} \right]$$

Applying the z -transform, we obtain

$$KG(z) = \frac{10.5K(z - 0.9048)}{(z - 1)^2}$$

The loci originate at $z = 1$ and terminate at $z = 0.9048$ and $z = -\infty$. There is one asymptote at 180° . The root locus is shown in Figure 7-10. The system becomes unstable when the closed-loop pole leaves the interior of the unit circle at point A shown in Figure 7-10. The value of K at this point can be determined from the condition $KG(z) = -1$. Therefore,

$$\left. \frac{10.5K(z - 0.9048)}{(z - 1)^2} \right|_{z = -1} = \frac{10.5K(-1.9048)}{4} = -1$$

Thus $K = 0.2$, and we see that the system is stable of $0 < K < 0.2$.

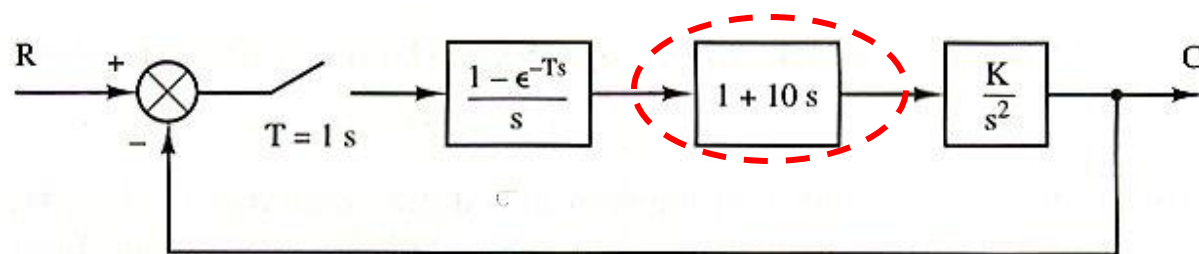
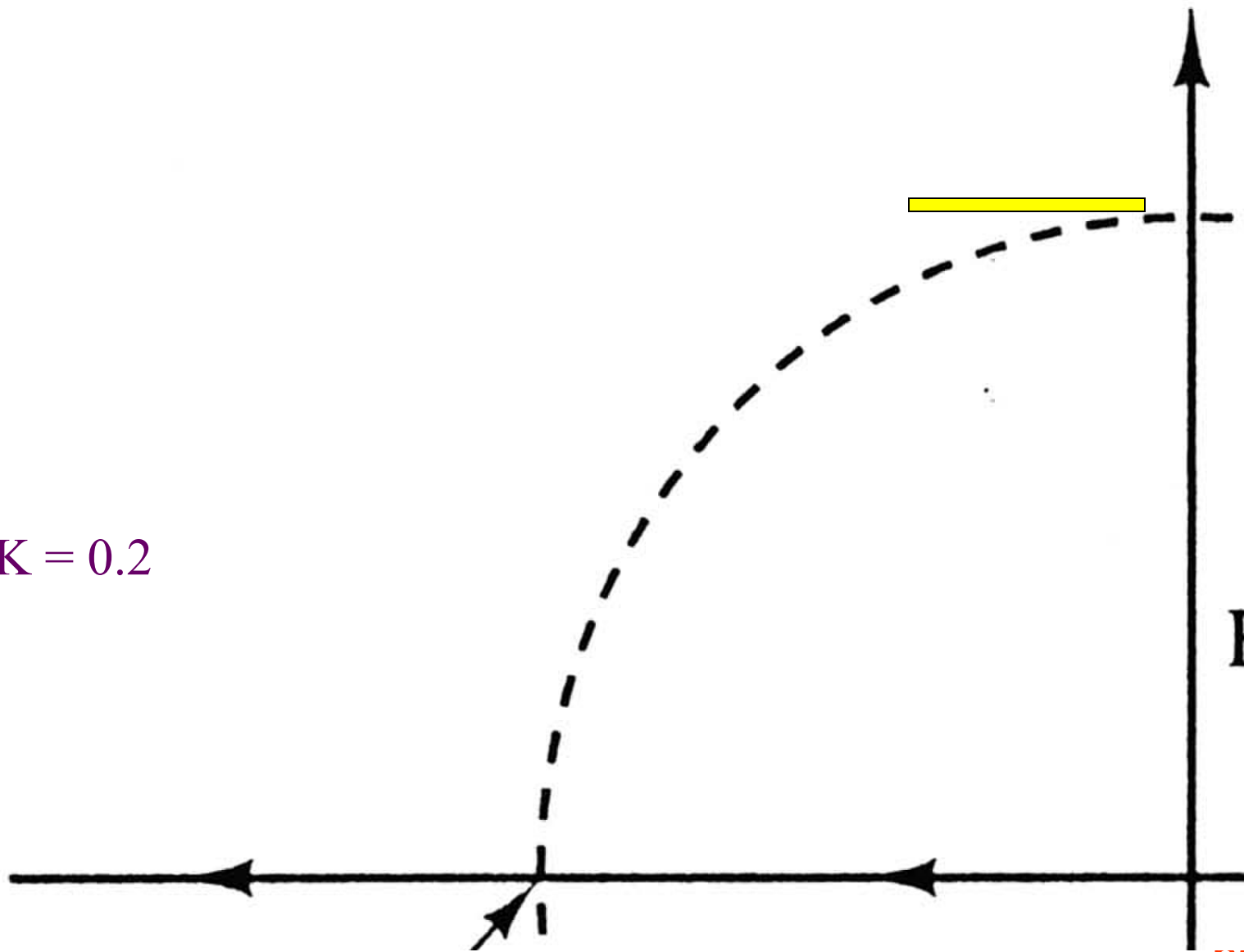


Figure 7-9 System for Example 7.8.

$K = 0.2$



7.7 The Nyquist Criterion

The transfer function for the continuous-time system of figure 7-11a is

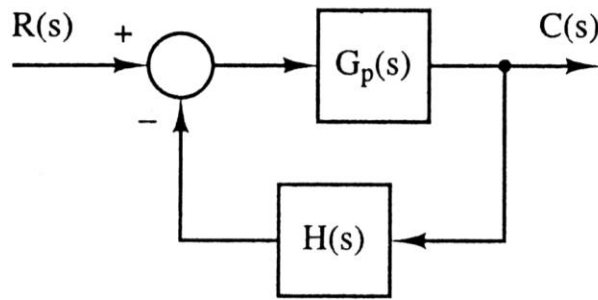
$$\frac{C(s)}{R(s)} = \frac{G_p(s)}{1 + G_p(s)H(s)} \quad (7.17)$$

And for the sampled-data system of Figure 7-11b the transfer function is

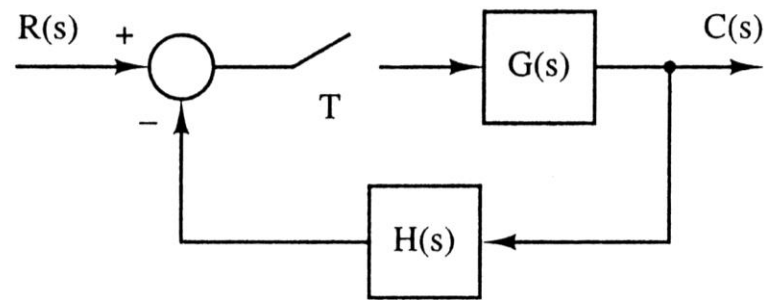
$$\frac{C^*(s)}{R^*(s)} = \frac{G^*(s)}{1 + \overline{GH}^*(s)} \quad (7.18)$$

The characteristic equation for the continuous-time system is $1 + G_p(s)H(s) = 0$ (7.19)

For the sampled-data system, the characteristic equation is $1 + \overline{GH}^*(s) = 0$ (7.20)



(a)



(b)

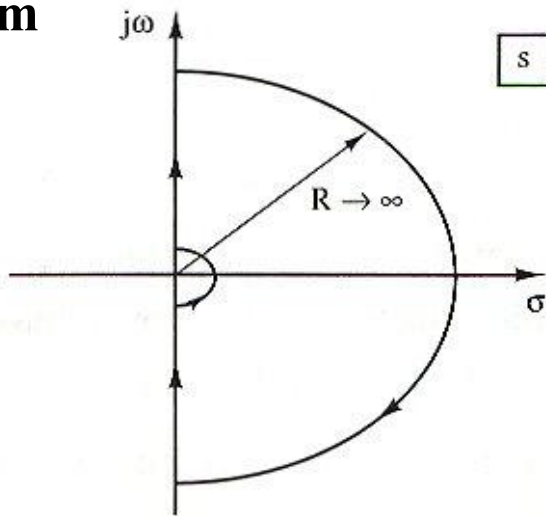
Figure 7-11 Continuous-time and sampled-data systems.

The characteristic equation of the sampled-data system can also be written as

$$1 + \overline{GH}(z) = 0 \quad (7.21)$$

Recall that the continuous-time system is stable if the roots of $1 + G_p(s)H(s) = 0$ are all contained in the left half-plane. similarly, the sampled-data system is stable if the roots of $1 + \overline{GH}^*(s) = 0$ all lie in the left half-plane, or if the roots of $1 + \overline{GH}(z) = 0$ all lie within the unit circle.

Theorem



$$N = Z - P \quad (7.22)$$

Phillips's textbook
N: clockwise

Figure 7-12 Nyquist path in the s-plane

10.3 INTRODUCTION TO THE NYQUIST CRITERION

- DERIVATION OF THE NYQUIST CRITERION

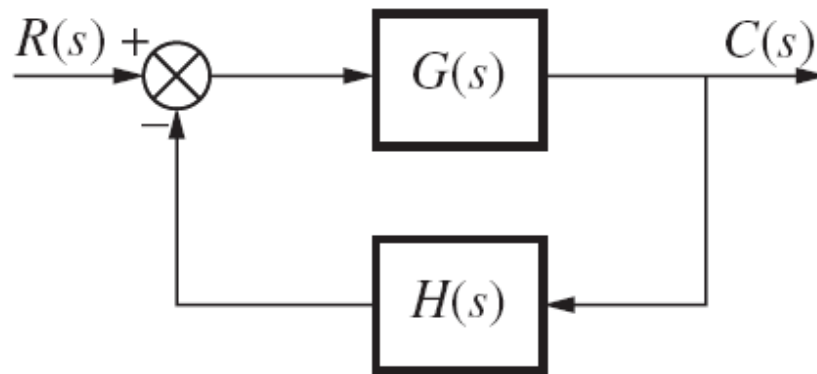


FIGURE 10.20 Closed-loop control system

GH (OL), 1+GH, and T (CL)

$$G(s) = \frac{N_G}{D_G} \quad (10.37a)$$

$$H(s) = \frac{N_H}{D_H} \quad (10.37b)$$

1+GH and GH (OL)
Poles are the same.

$$G(s)H(s) = \frac{N_G N_H}{D_G D_H} \quad (10.38a)$$

★ $1 + G(s)H(s) = 1 + \frac{N_G N_H}{D_G D_H} = \frac{D_G D_H + N_G N_H}{D_G D_H} \quad (10.38b)$

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{N_G D_H}{D_G D_H + N_G N_H} \quad (10.38c)$$

1+GH Zeros
= CL Poles

Mapping Points & Contours

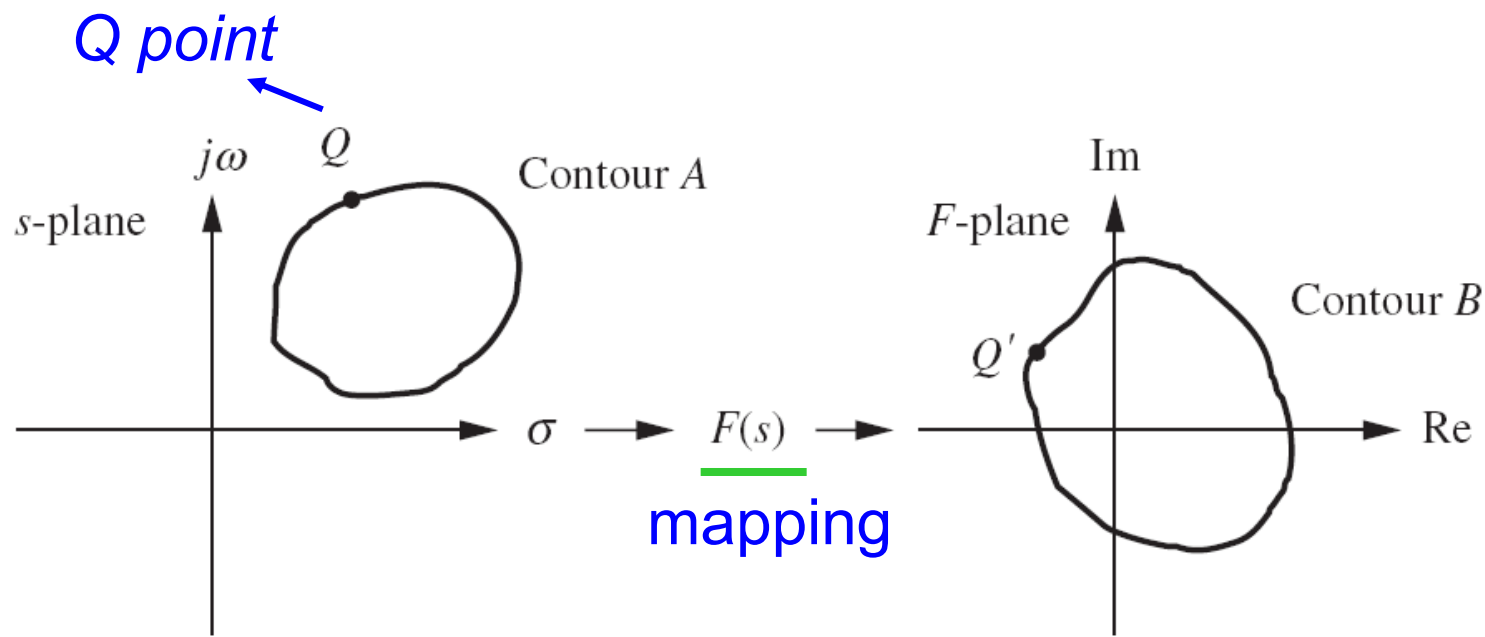


FIGURE 10.21 Mapping contour A through function $F(s)$ to contour B

$$F(s) = \frac{(s - z_1)(s - z_2) \dots}{(s - p_1)(s - p_2) \dots} \quad (10.39)$$

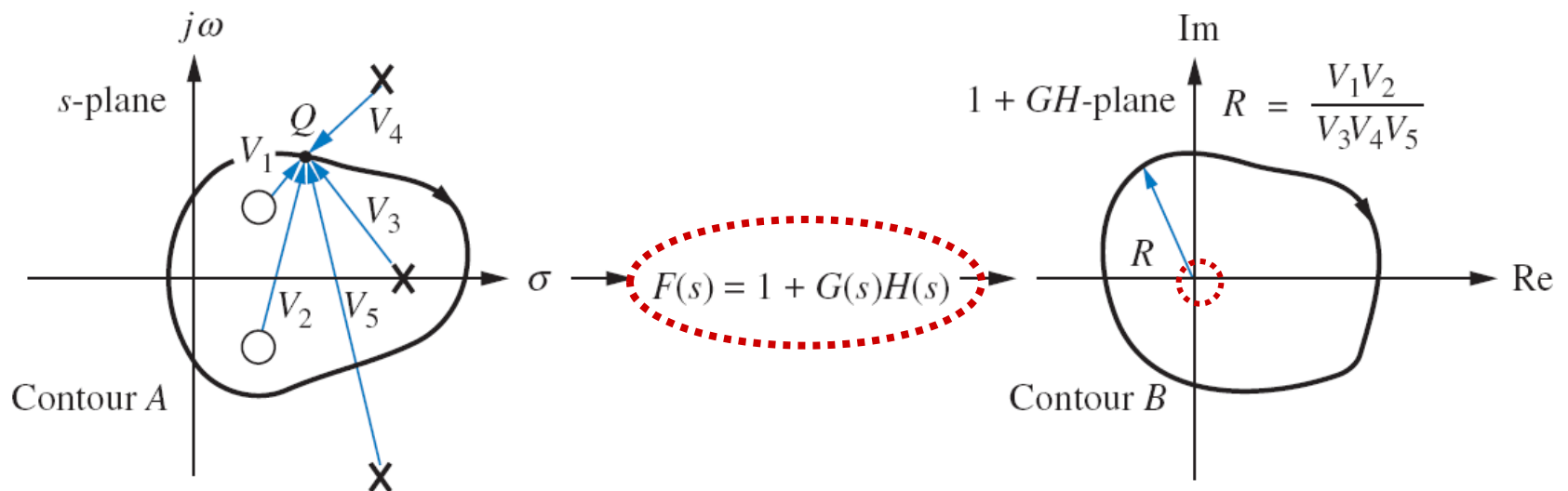
Nyquist Criterion

Nise's textbook
N: counterclockwise

$$F(s) = \frac{(s - z_1)(s - z_2) \dots}{(s - p_1)(s - p_2) \dots} \quad (10.39)$$

FIGURE 10.23 Vector representation of mapping

- Inside Contour A (move clockwise)
 - Zero: clockwise
 - Pole: counterclockwise
- $N = P - Z$, N : CCW around origin



Nyquist Plot of GH

$F(s)=1+GH$, inside Contour A:
 $Z = P - N$,
where N is CCW around origin

What if $F(s)=GH$?

$Z \rightarrow$ CL poles in rhp (unknown)

$P \rightarrow$ OL poles in rhp (given)

$N \rightarrow$ CCW around -1 (found)

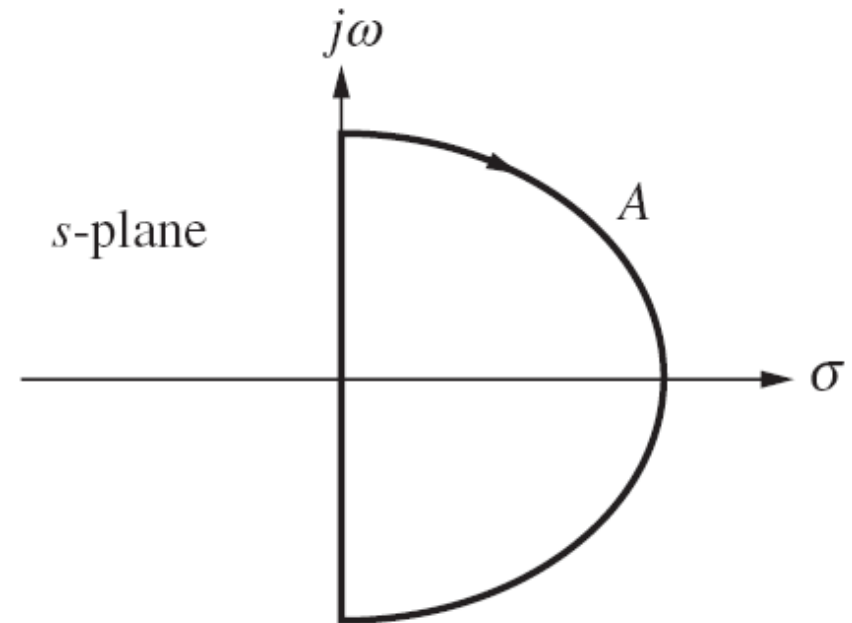


FIGURE 10.24 Contour enclosing right half-plane to determine stability

Supplementary Materials

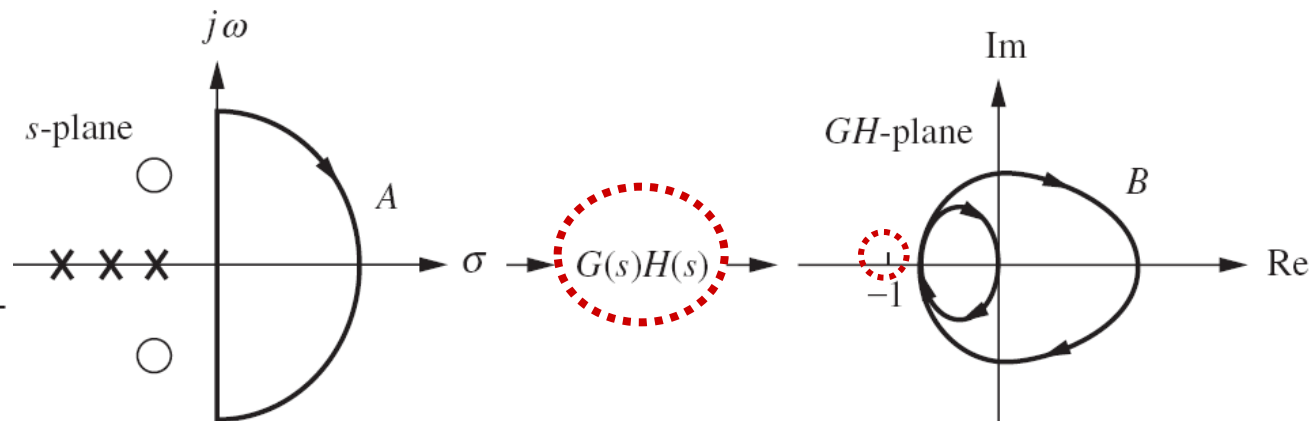
APPLYING THE NYQUIST CRITERION TO DETERMINE STABILITY

FIGURE 10.25

Mapping examples:

a. Contour does not enclose closed-loop poles;

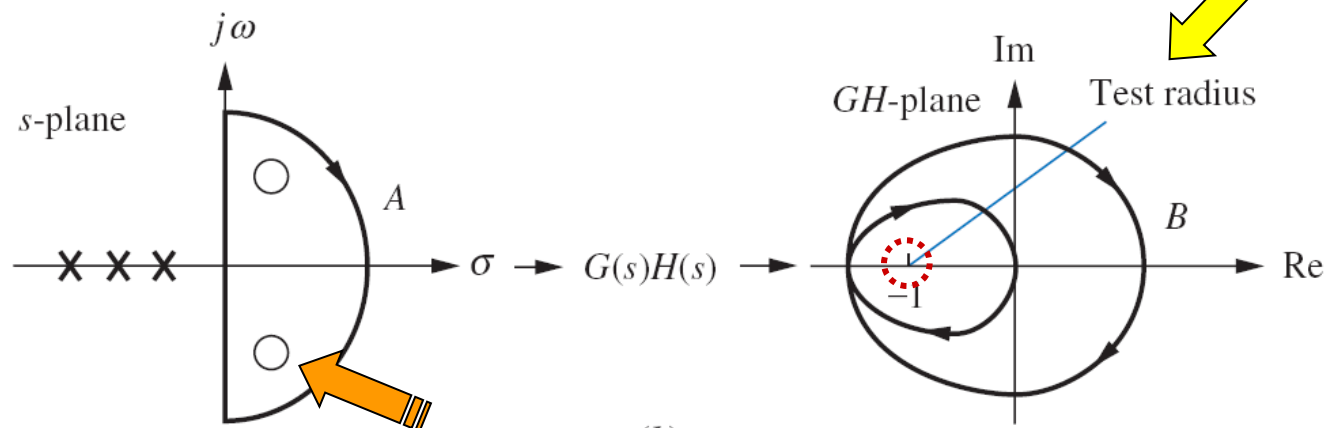
b. contour does enclose closed-loop poles



(a)

$$\begin{aligned} (a) \quad Z &= P - N \\ &= 0 - 0 = 0 \end{aligned}$$

➡ Stable!



(b)

$$\begin{aligned} (b) \quad Z &= P - N \\ &= 0 - (-2) = 2 \end{aligned}$$

➡ Unstable!

○ = zeros of $1 + G(s)H(s)$
= poles of closed-loop system
Location not known

✕ = poles of $1 + G(s)H(s)$
= poles of $G(s)H(s)$
Location is known

Example 7.9

S-plane

As an example of the Nyquist criterion applied to a continuous-time system, consider the system of Figure 7-13. The Nyquist path and the resultant Nyquist diagram are shown in Figure 7-14. The small detour taken by the Nyquist path around the origin, labeled I on Figure 7-14a, is necessary, since $G_p(s)$ has a pole at the origin. Along this detour, let

$$s = \rho e^{j\theta}, \quad \rho \ll 1$$

Then

$$G_p(s)|_{s = \rho e^{j\theta}} \approx \frac{K}{\rho e^{j\theta}} = \frac{K}{\rho} \angle -\theta$$

Thus the small detour generates the large arc on the Nyquist diagram. For the Nyquist path along the $j\omega$ -axis (II),

$$G_p(s)|_{s = j\omega} = \frac{K}{j\omega(1 + j\omega)} = \frac{K}{\omega\sqrt{1 + \omega^2}} \angle -90^\circ - \tan^{-1} \omega$$

This portion of the Nyquist diagram is simply the frequency response of the system. Along the large arc of the Nyquist path, $G_p(s) \approx 0$. The complete Nyquist diagram is given in Figure 7-14b. Now let us apply the Nyquist criterion. We can see from the Nyquist diagram in Figure 7-14 that $N = 0$. Since the open-loop transfer function has no poles in the right half-plane, $P = 0$, and therefore from the equation

$$Z = N + P$$

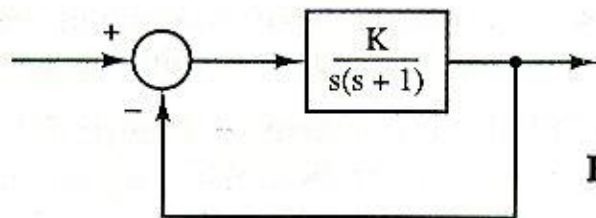
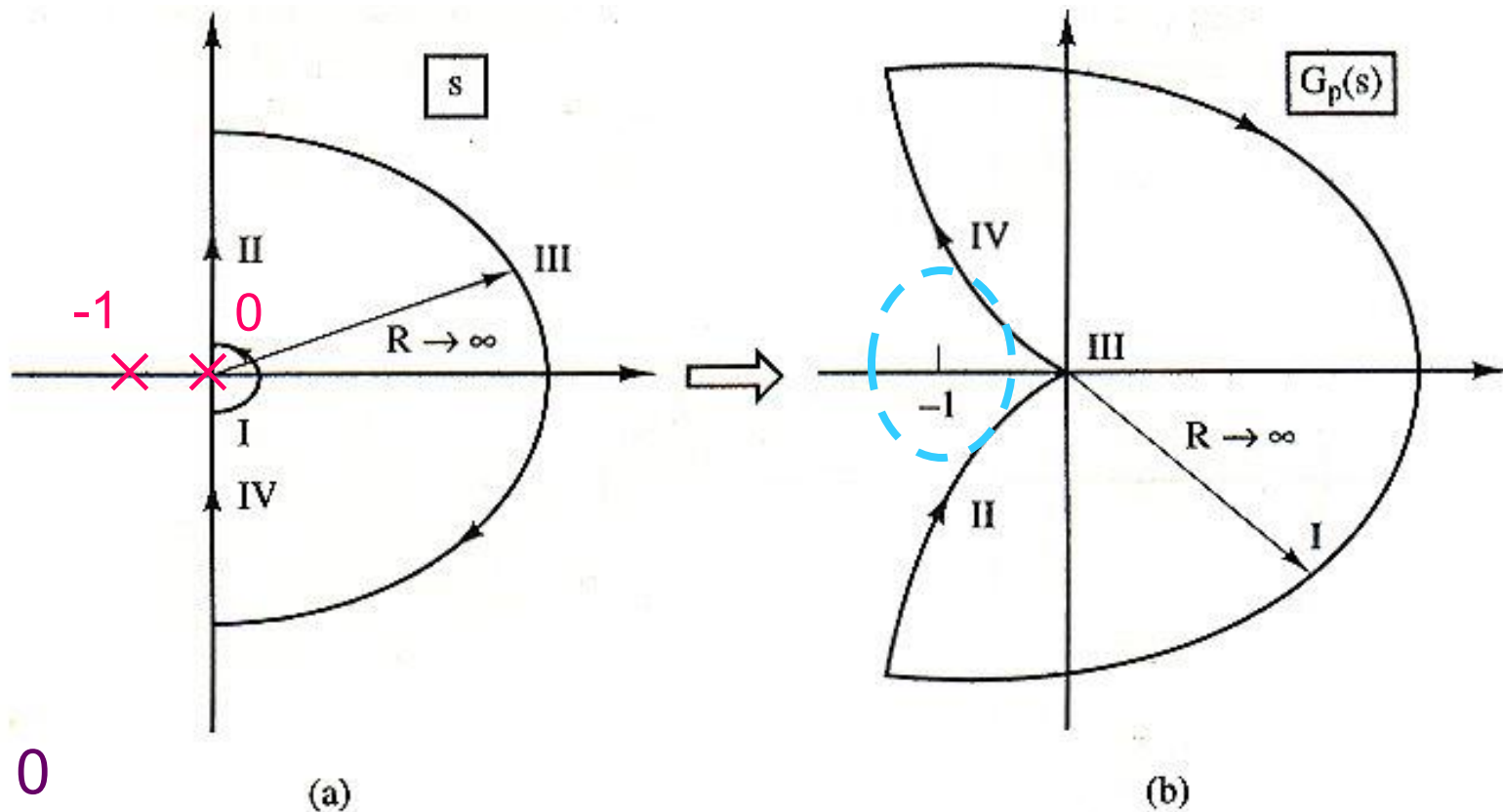


Figure 7-13 System for Example 7.9.

we find that $Z = 0$. Thus the system is stable. In addition, from the Nyquist diagram in Figure 7-14 it can be seen that the system is stable for all $K > 0$.



$Z = 0$

➡ Stable!

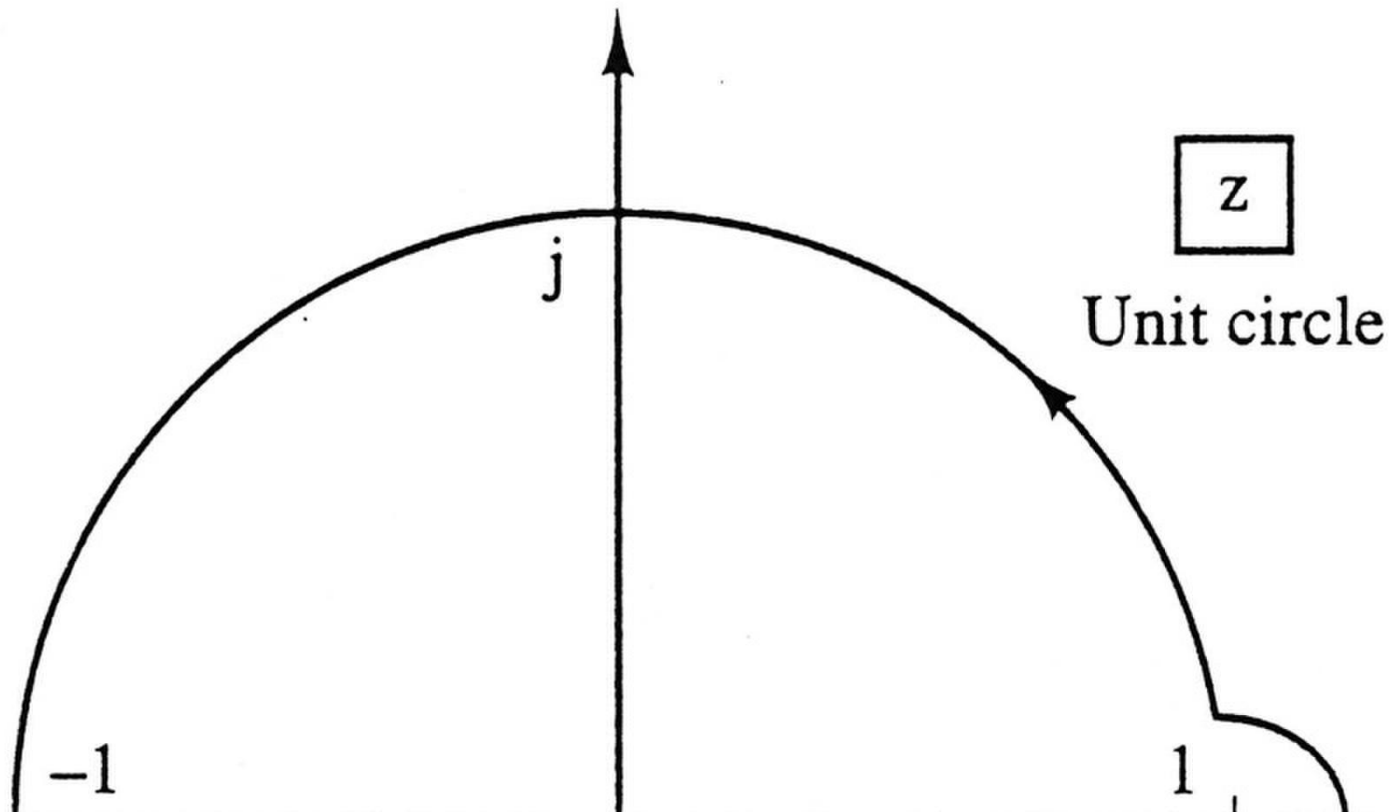
Figure 7-14 Nyquist diagram for Example 7.9.

$$N = Z - P \quad (7.28)$$

N = number of **clockwise encirclements** of the -1 point

Z = number of zeros of the characteristic equation
outside the unit circle

P = number of poles of the open-loop function,
or **equivalently poles of the characteristic equation**, that are **outside the unit circle**



Example 7.10

The Nyquist criterion will now be illustrated using the system of Figure 7-16. From Example 6.4,

$$G(z) = \frac{0.368z + 0.264}{(z - 1)(z - 0.368)}$$

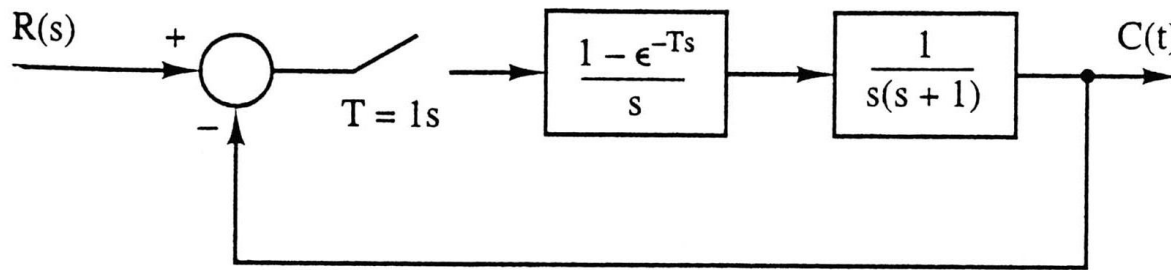


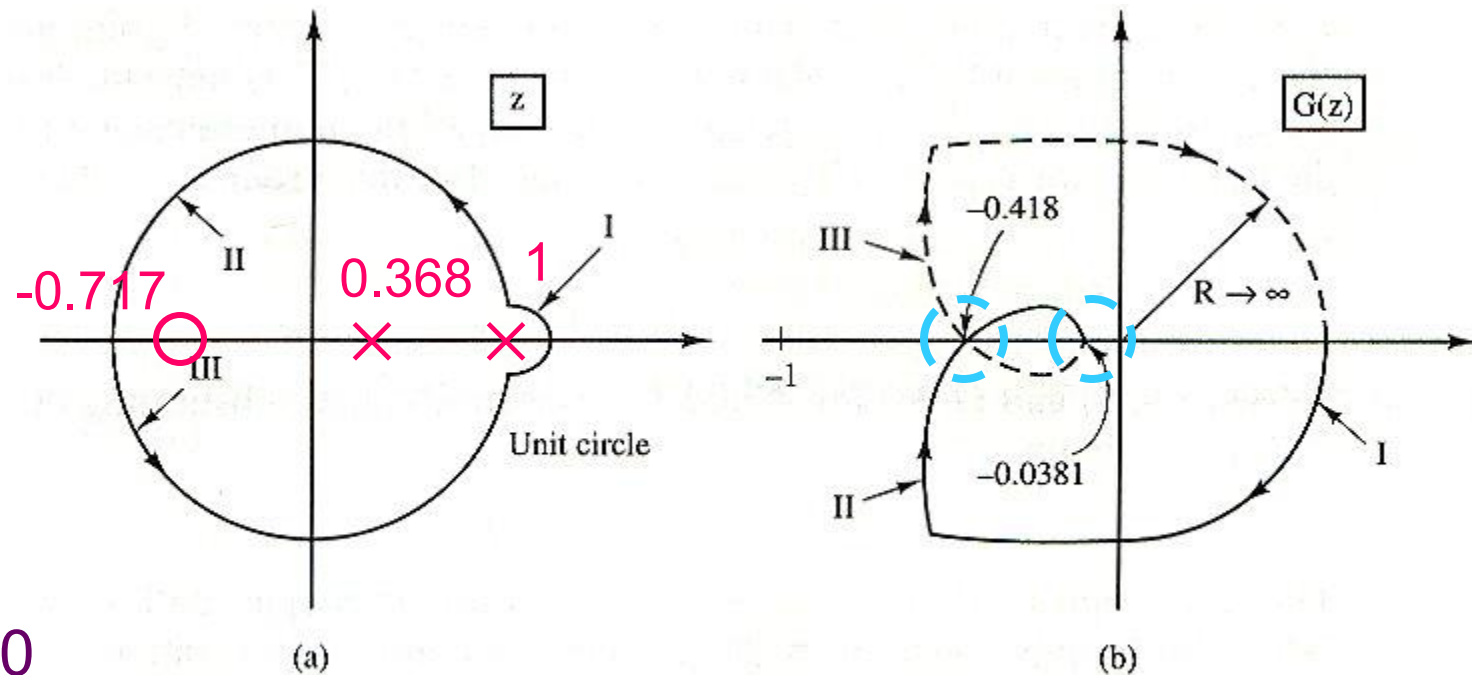
Figure 7-16 System for Example 7.10.

The Nyquist path and the Nyquist diagram are both shown in Figure 7-17. The detour around the $z = 1$ point on the Nyquist path (Figure 7-17a) is necessary since $G(z)$ has a pole at this point. On this detour,

$$z = 1 + \rho e^{j\theta}, \quad \rho \ll 1$$

and

$$G(z)|_{z=1+\rho e^{j\theta}} \approx \frac{0.632}{\rho e^{j\theta}(0.632)} = \frac{1}{\rho} \angle -\theta$$



$Z = 0$

➡ Stable!

Figure 7-17 Nyquist path and Nyquist diagram for Example 7.10.

Thus this detour generates the large arc on the Nyquist diagram (Figure 7-17b). For z on the unit circle,

$$G(z)|_{z=e^{j\omega T}} = \frac{0.368e^{j\omega T} + 0.264}{(e^{j\omega T} - 1)(e^{j\omega T} - 0.368)}$$

In this equation ω varies from $-\omega_s/2$ to $\omega_s/2$. Since $G(e^{j\omega T})$ for $0 > \omega > -\omega_s/2$ is the complex conjugate of $G(e^{j\omega T})$ for $0 < \omega < \omega_s/2$, it is necessary to calculate $G(e^{j\omega T})$ only for $0 < \omega < \omega_s/2$. This calculation results in the frequency response for $G(z)$. Note that $G(-1) = -0.0381$.

The calculation of the Nyquist diagram for the portion of the diagram labeled II in Figure 7-17b may be accomplished in several ways. We may use

$$G^*(s)|_{s=j\omega} \quad \text{or} \quad G(z)|_{z=e^{j\omega T}}$$

as the basis for calculation. Generally, computer calculations are required to evaluate the frequency response.

We see from Figure 7-17 that $N = 0$, since the Nyquist diagram does not encircle the -1 point. In addition, $P = 0$ since $G(z)$ has no poles outside the Nyquist path. Hence $Z = (N + P) = 0$, and the system is stable.

Suppose that a gain K is added to the plant. The system can be forced into instability by increasing this gain K by a factor of $1/0.418$, or 2.39 . However, the same system without sampling, as shown in Example 7.9, is stable for all positive values of gain K . As stated earlier, the destabilizing effect of the sampling can be seen from the phase lag introduced by the sampler and data hold.

“Phase lag” leads to phase margin $\Phi_M \downarrow$.

Table 7-4 gives the functions and the range of variables required for the calculation of the Nyquist path

TABLE 7-4 FUNCTIONS NYQUIST DIAGRAM

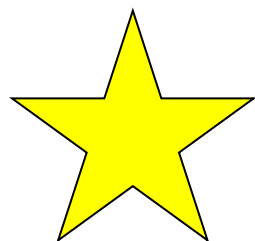
Open-loop function

Example 7.9

Example 7.10

Example 7.11

10.6 GAIN MARGIN AND PHASE MARGIN



Assume $N = 0$ for stability
→ $P = 0$ (no OL poles in rhp)

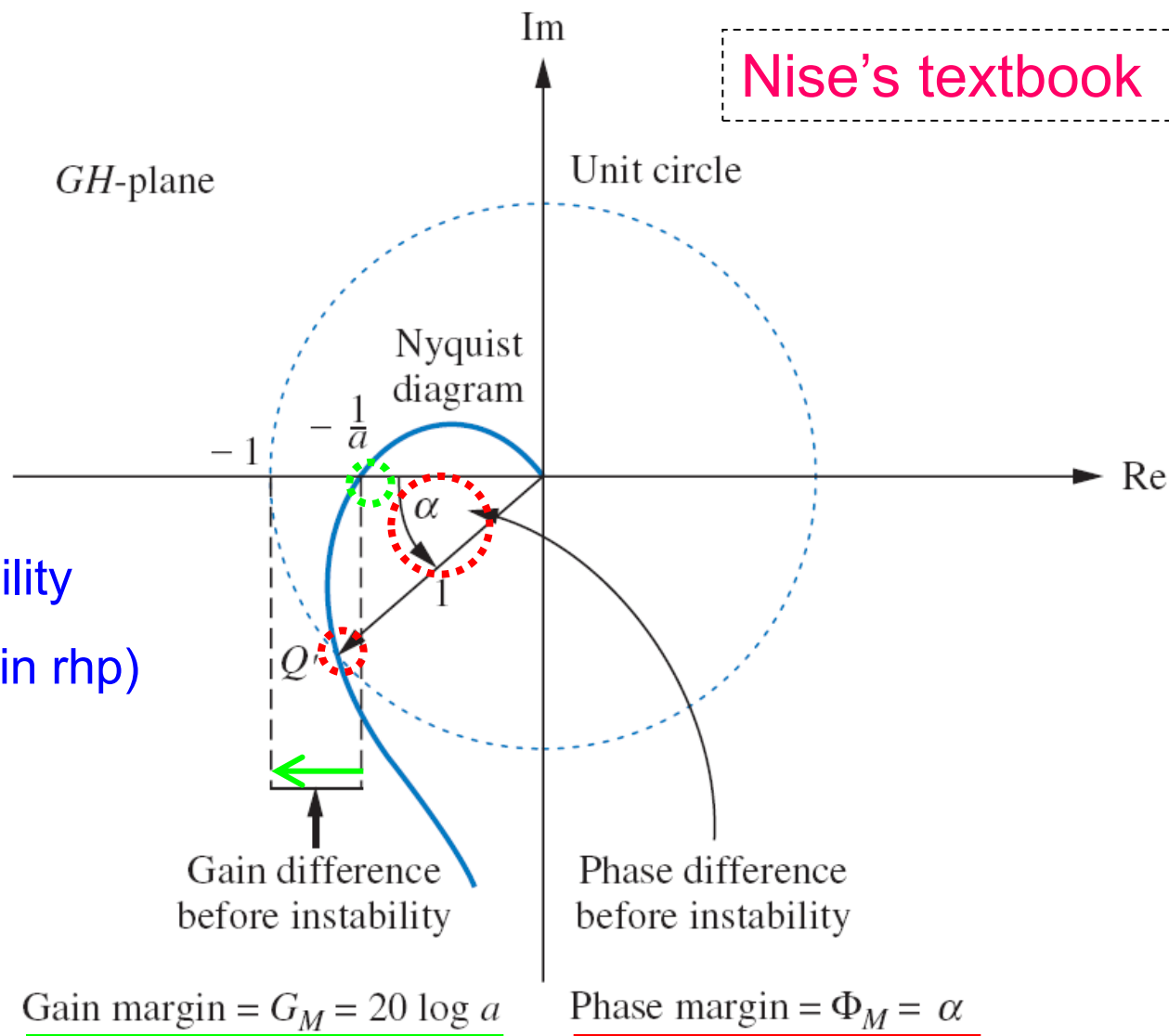
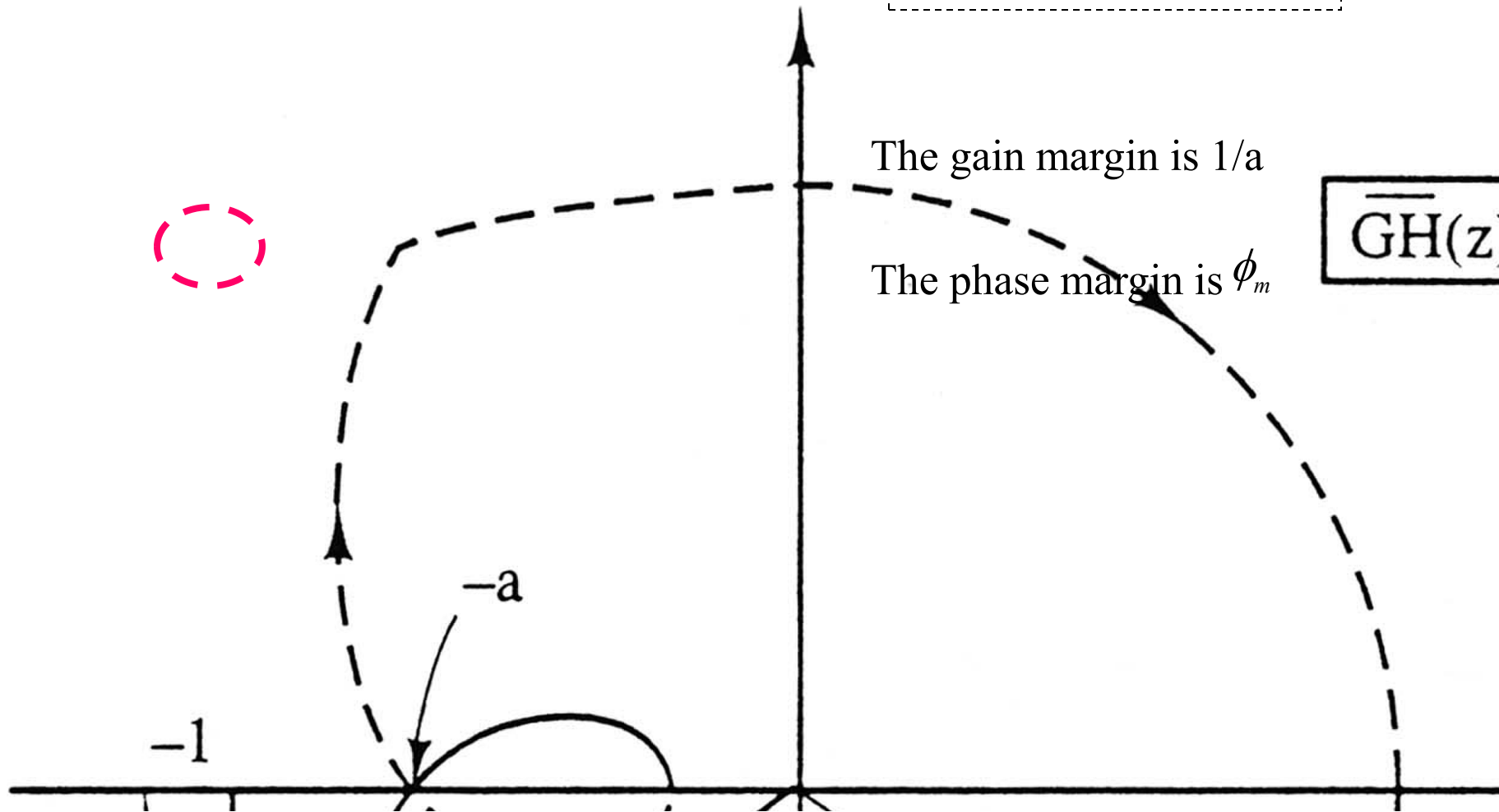


FIGURE 10.35 Nyquist diagram showing gain and phase margins

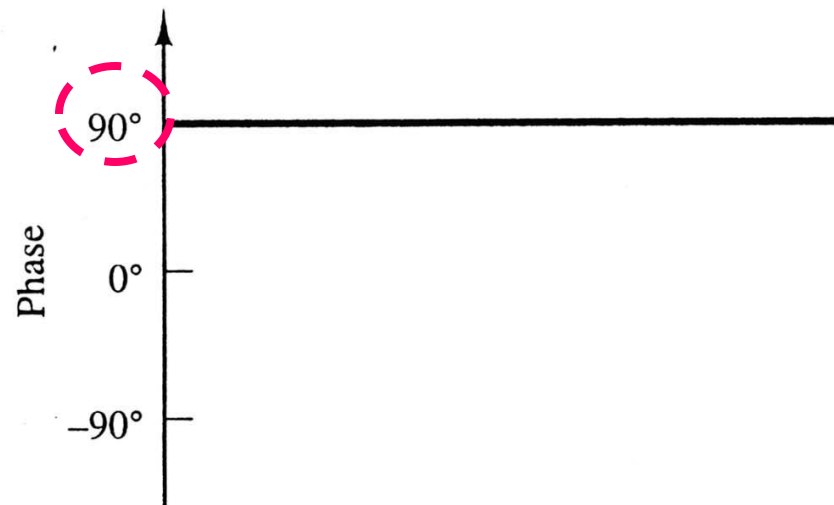
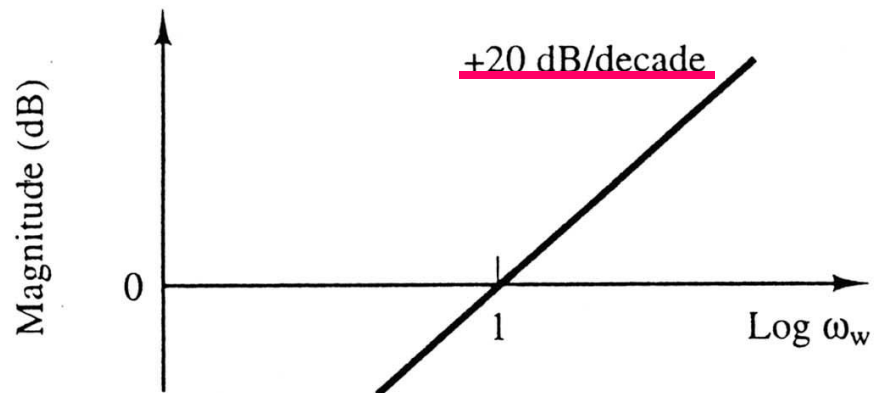
Phillips's textbook



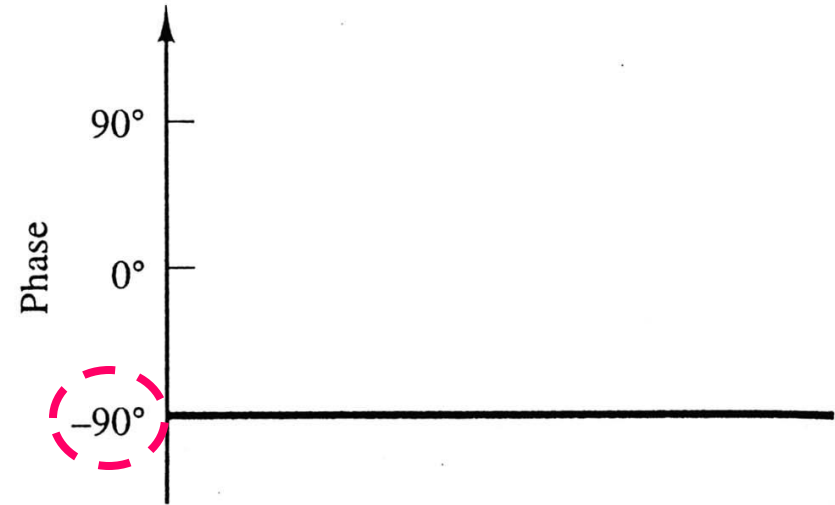
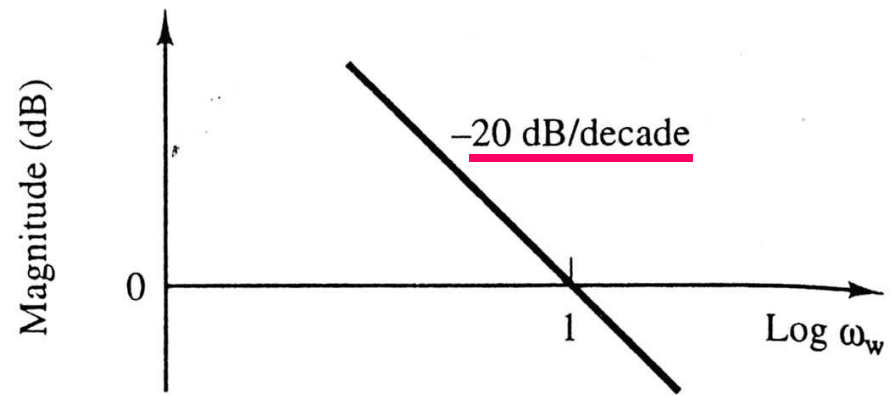
7.8 The Bode Diagram

TABLE 7-5 SUMMARY OF TERMS EMPLOYED IN A BODE DIAGRAM

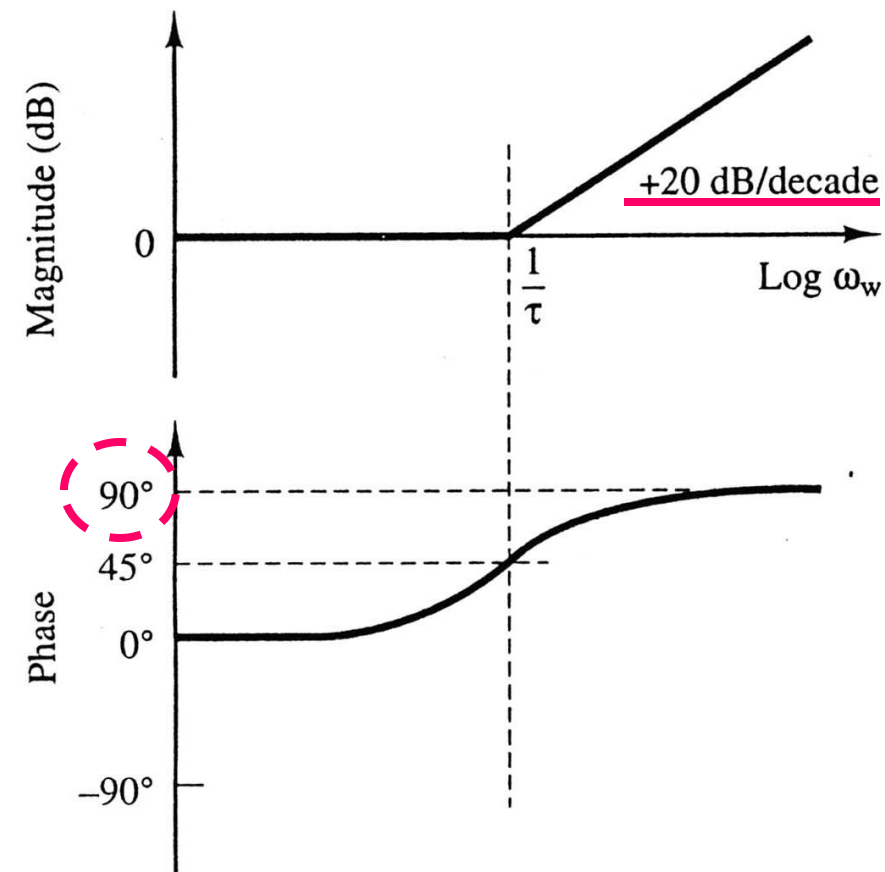
1. A constant term K . When this term is present, the log magnitude plot is shifted up or down by the amount $20 \log_{10} K$.
2. The term $j\omega_w$ or $1/j\omega_w$. If the term $j\omega_w$ is present, the log magnitude is $20 \log_{10} \omega_w$, which is a straight line with a slope of $+20$ dB/decade, and the phase is constant at $+90^\circ$. If the term $1/j\omega_w$ is present, the log magnitude is $-20 \log \omega_w$, which is a straight line with a slope of -20 dB/decade, and the phase is constant at -90° . The Bode plots for these terms are shown in Figure 7-20.
3. The term $(1 + j\omega_w \tau)$ or $[1/(1 + j\omega_w \tau)]$. The term $(1 + j\omega_w \tau)$ has a log magnitude of $20 \log_{10} \sqrt{1 + \omega_w^2 \tau^2}$ which can be approximated as $20 \log_{10} 1 = 0$ when $\omega_w \tau \ll 1$ and as $20 \log_{10} \omega_w \tau$, $\omega_w \tau \gg 1$. The corner or “break” frequency is $\omega_w = 1/\tau$. The phase is given by the expression $\tan^{-1} \omega_w \tau$. The term $[1/(1 + j\omega_w \tau)]$ is handled in a similar manner. The Bode plots for these functions are shown in Figure 7-20.



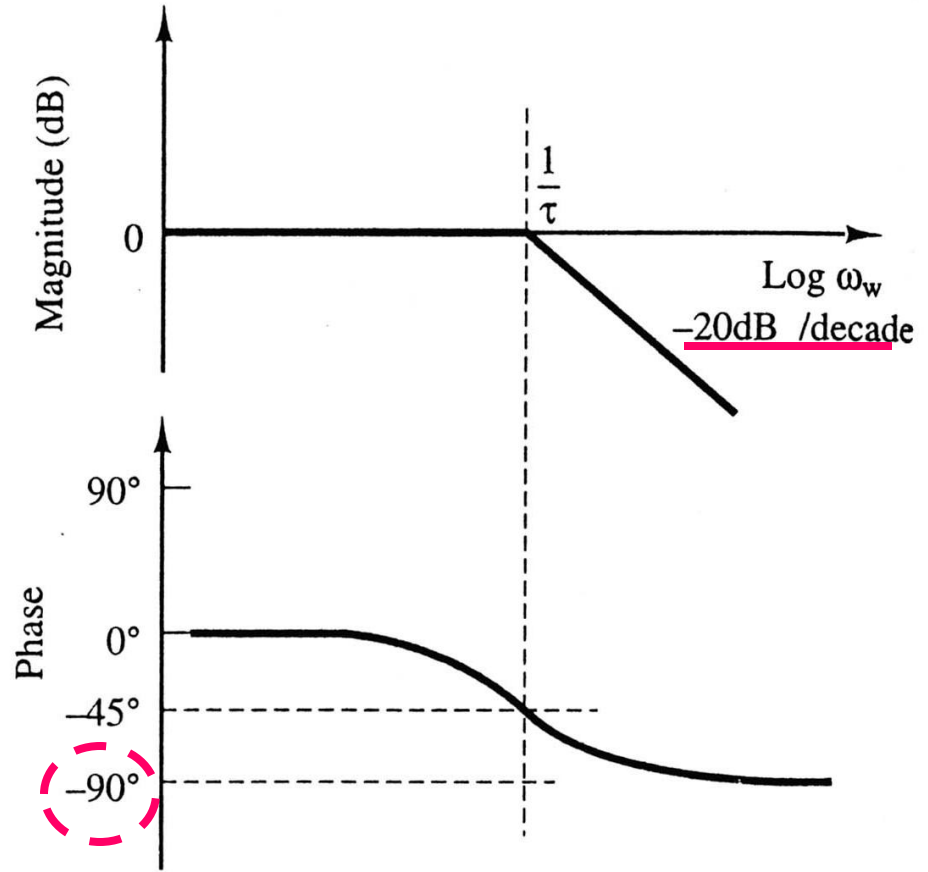
(a)



(b)



(c)



(d)

Figure 7-20 Short summary of terms employed in Bode diagrams: (a) Bode plot for $j\omega_w$; (b) Bode plot for $1/j\omega_w$; (c) Bode plot for $1 + j\omega_w\tau$; (d) Bode plot for $1/(1 + j\omega_w\tau)$.

Example 7.12

Consider again the system of Example 7.3. For this system,

$$G(w) = -\frac{0.0381(w - 2)(w + 12.14)}{w(w + 0.924)}$$

and

$$\begin{aligned} G(j\omega_w) &= -\frac{0.0381(j\omega_w - 2)(j\omega_w + 12.14)}{j\omega_w(j\omega_w + 0.924)} \\ &= \frac{-\left(j\frac{\omega_w}{2} - 1\right)\left(\frac{j\omega_w}{12.14} + 1\right)}{j\omega_w\left(\frac{j\omega_w}{0.924} + 1\right)} \end{aligned}$$

Note that the numerator break frequencies are $\omega_w = 2$ and $\omega_w = 12.14$ and the denominator break frequencies are $\omega_w = 0$ and $\omega_w = 0.924$. The Bode diagram for this system, using straight-line approximations, is shown in Figure 7-21. Both the gain and phase margins of the system are shown on the diagram.

As stated in Example 7-10, this system can be made unstable by increasing the gain. This can also be seen in Figure 7-21. Increasing the gain is equivalent to shifting the entire magnitude curve vertically upward. Therefore, if the gain is increased by an amount equal to the gain margin, a condition of marginal stability will exist. A

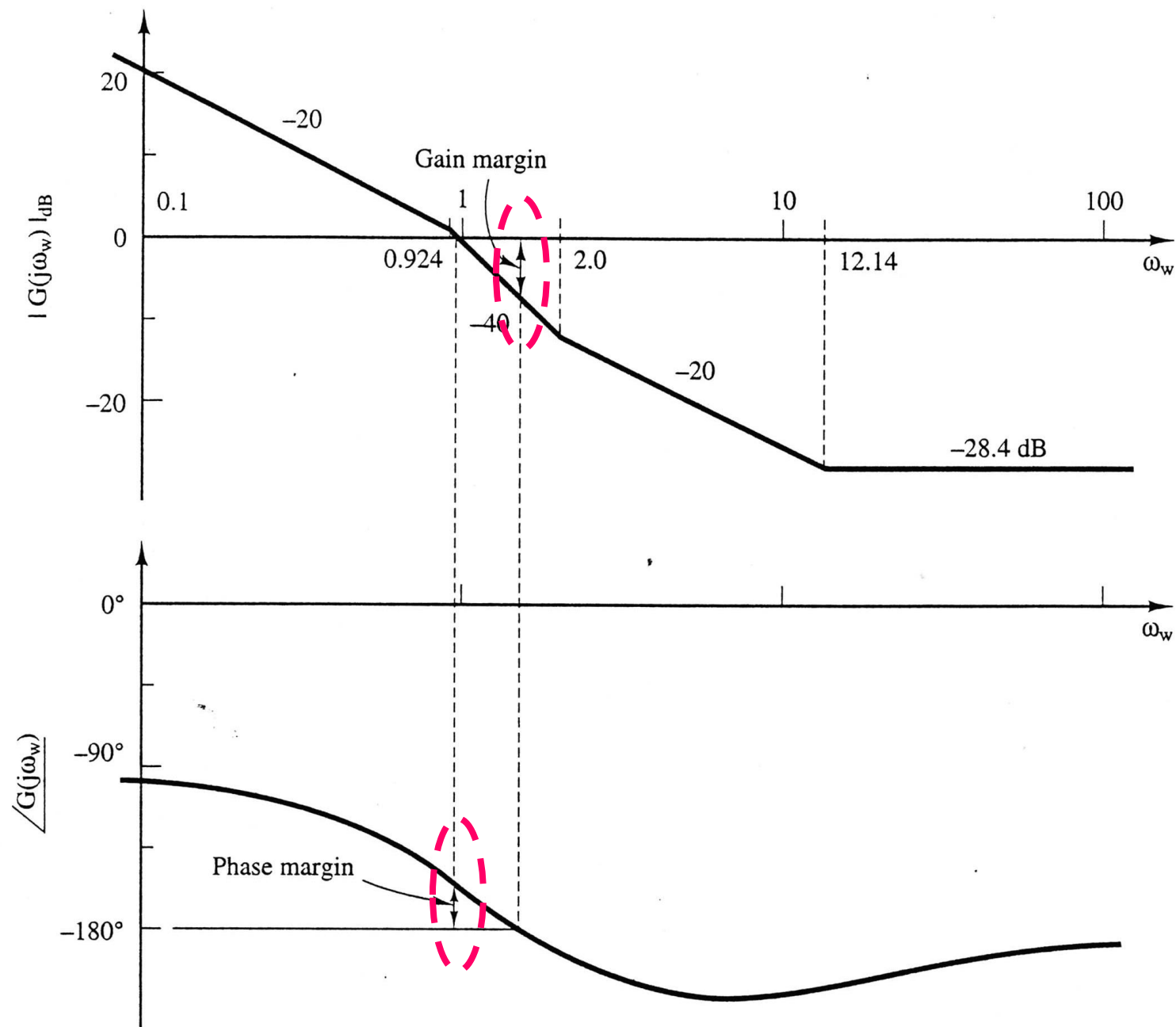
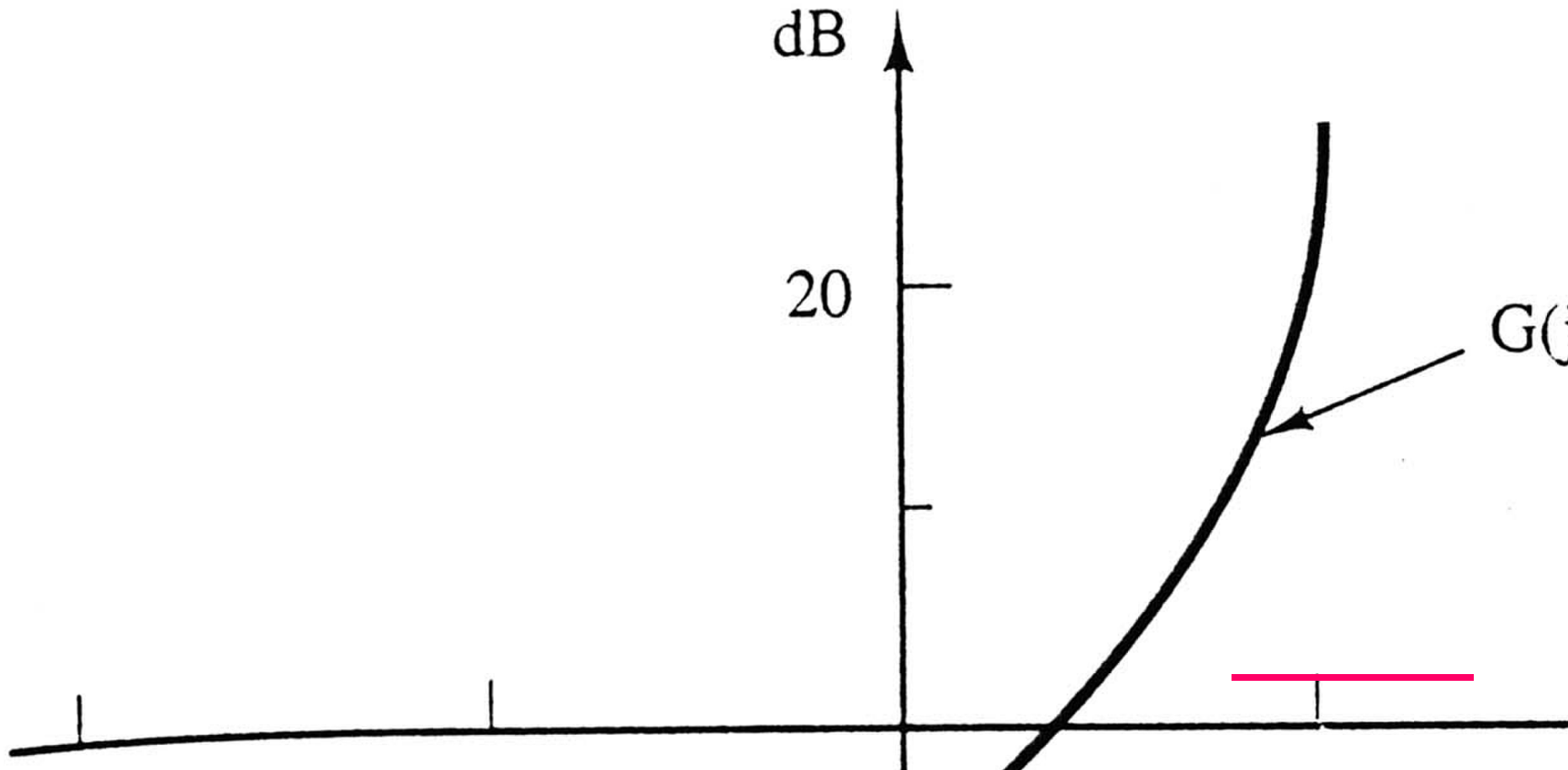
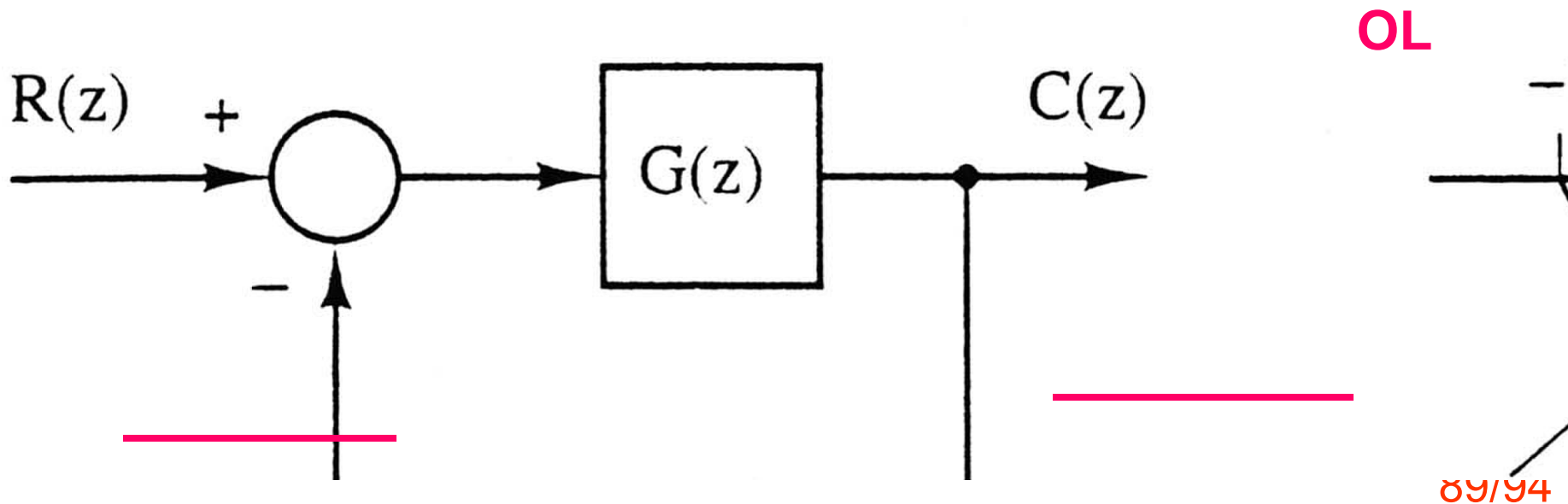


Figure 7-21 Bode diagram for Example 7.12.

7.9 Interpretation of the Frequency Response



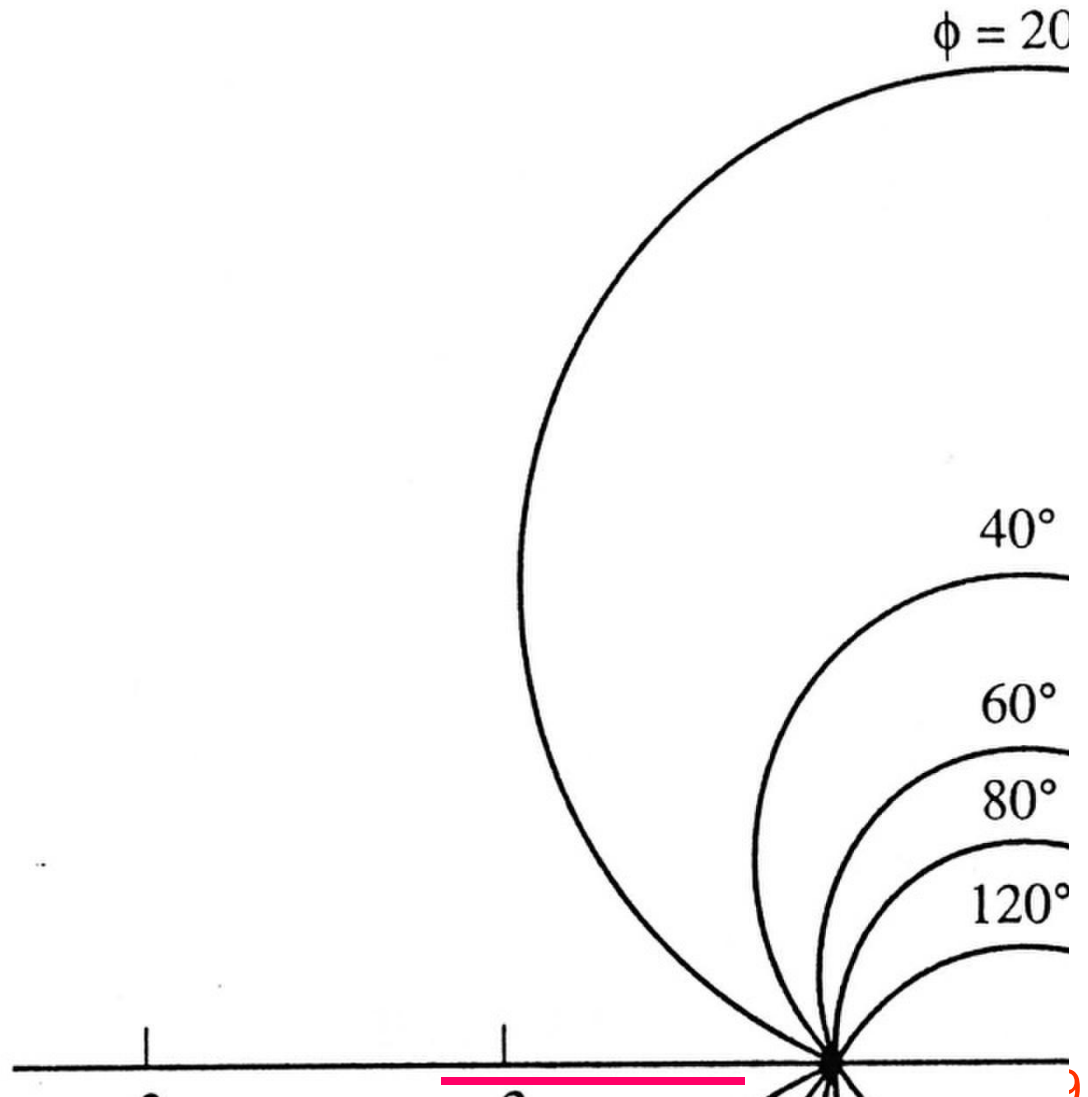
7.10 Closed-Loop Frequency Response



Constant M circles:

The locus of the CL $Magnitude$ FR

Constant N circles:
The locus of the CL $Phase$ FR



Nichols Chart: Gain-Phase Plot (M in dB)

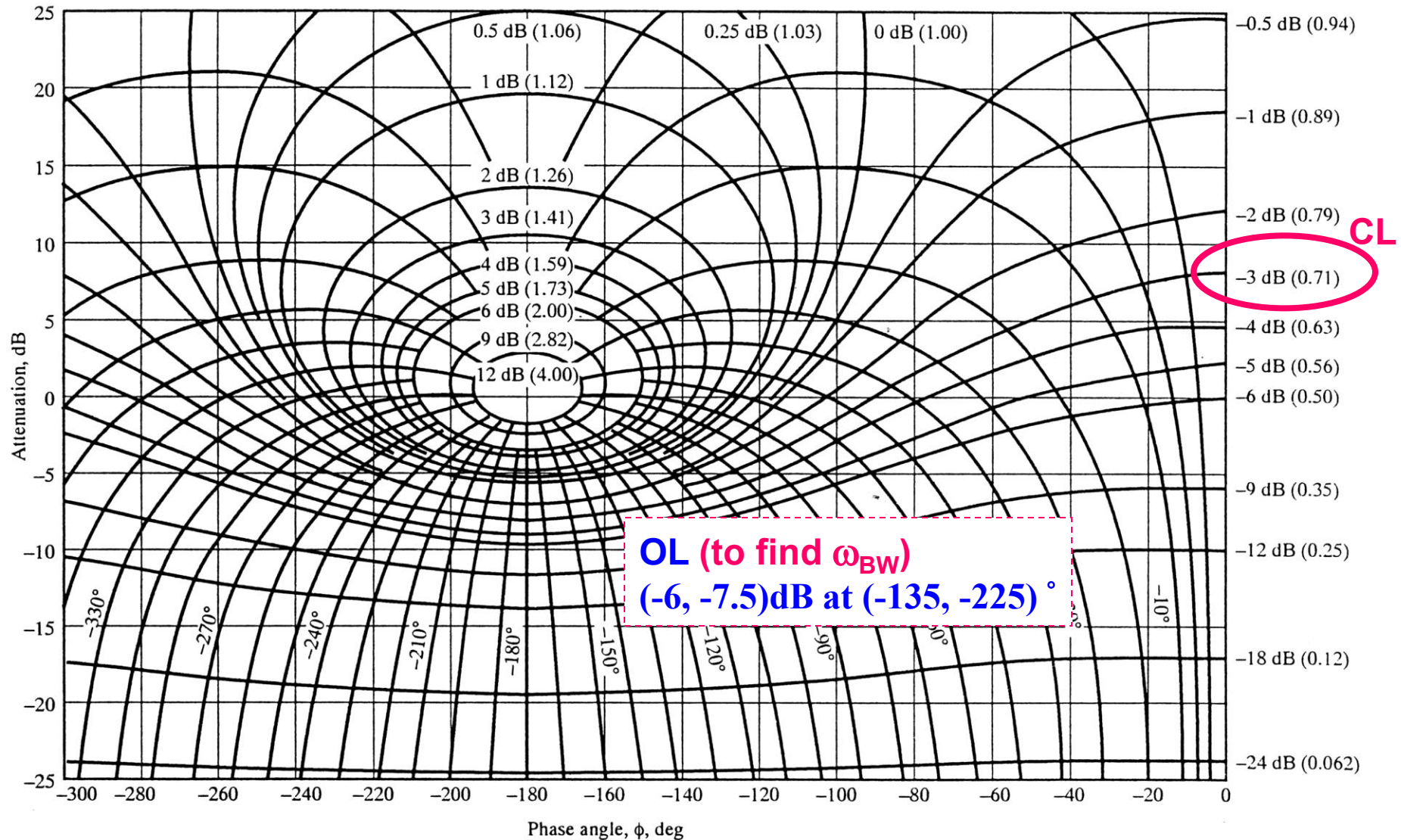


Figure 7-26 Nichols chart.

Example 7.13

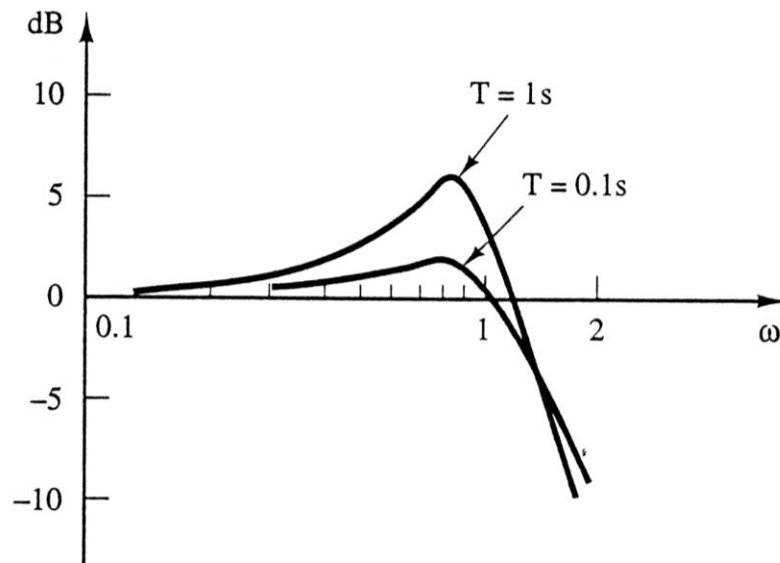


Figure 7-27 Closed-loop frequency response magnitudes for Example 7.13.

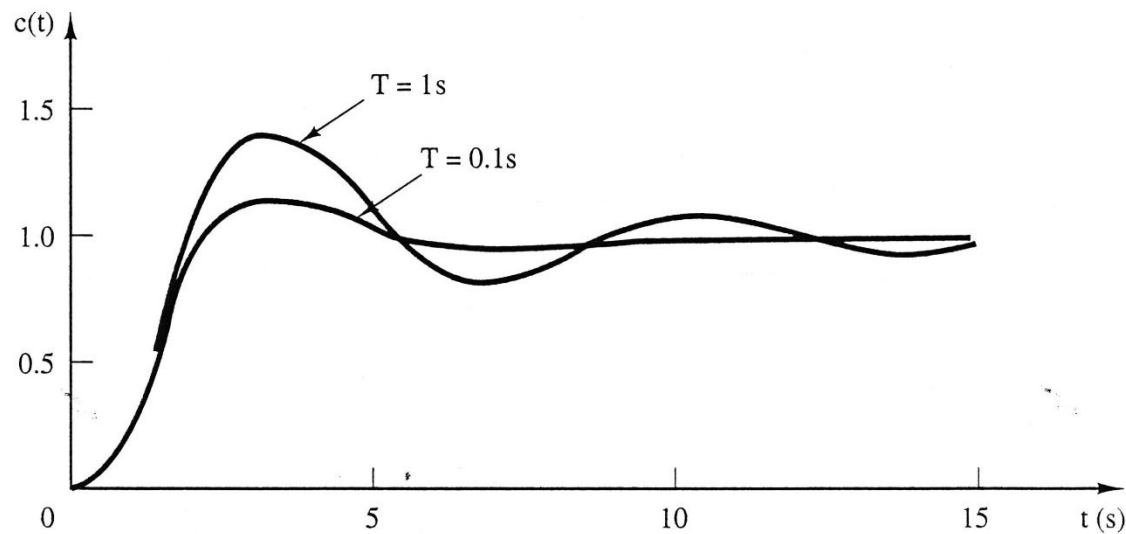
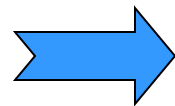


Figure 7-28 Step responses for Example 7.13.

- See Summary (P272)
- Problems: 7, 8, 13, 14, 19

 Test Bank